Parametric Constructions, Quadratic Norm Identities, and Birational Geometry of the Diophantine Equation $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$ over $\mathbb{Z}\setminus\{0\}$

October 2026

Abstract

This paper presents an algebraic, number-theoretic, and geometric investigation of primitive non-zero integer solutions to the fourth-power Diophantine equation

$$a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4 \quad \text{over } \mathbb{Z}\setminus\{0\}.$$

Although the smooth projective quartic threefold $V\subset\mathbb{P}^4$ defined by this equation bounds a strictly convex compact body in the affine chart $a+b+c+d+e=1$—and consequently contains no real straight lines—it admits a hierarchy of quadratic norm-form decompositions, bitangent hyperplane sections, conic bundles, and osculating rational maps. We first establish the strict compactness bound $1\le|a|,|b|,|c|,|d|,|e|\le\lfloor(s^4-4)^{1/4}\rfloor\le s-1$ (ruling out any unbounded cancellation region for $s=a+b+c+d+e>0$) together with the 2-Adic Parity Law proving that every primitive solution has odd sum s, a unique odd coordinate, and four even coordinates. Using two independent exhaustive enumerations over $0<s<1000$—a meet-in-the-middle reduction on the four even coordinates and a parity-unrestricted fourth-power-budget search over all $1\le s\le 999$ (with complete C99 source code in Appendix A)—we establish that the census of 42 primitive solutions with $0<s<1000$ (36 solutions with $5\mid s$, all satisfying $s\equiv 5 \pmod{10}$, and 6 solutions with $5\nmid s$, all satisfying $5^4\mid(s-a)$) is exhaustive. We prove that $V(\mathbb{C})$ is smooth and contains conjugate pairs of straight lines $L,\overline{L}$ over $\mathbb{Q}(\sqrt{-3})$ lying in the base plane $\Pi_0=\{a+b+c=d=0\}$ (where $V\cap\Pi_0=2L+2\overline{L}$), and that the hyperplanes H_{e_0} defined over $\mathbb{Q}$ are tangent to $V(\mathbb{C})$ at six points along $L\cup\overline{L}$ and slice V in quadratic conic bundles. For all four rational bitangent sectors present in the census ($e_0=-1,2,\frac{3}{5},\frac{1}{3}$), we construct explicit quadratic polynomial parametrizations that are birational (degree 1) onto their image curves over smooth genus-1 quartics, derive their explicit birational isomorphisms onto integral Weierstrass models $Y_W^2=X_W^3+a_2X_W^2+a_4X_W+a_6$ ($a_i\in\mathbb{Z}$), prove via the Nagell–Lutz Theorem that the base points have infinite order ($\operatorname{rank}E(\mathbb{Q})\ge 1$), prove zero-coordinate avoidance (at most 24 points with $abcdes=0$), and establish universal 5-adic sector invariance (showing that every rational point on $\mathcal{E},\mathcal{E}_2,\mathcal{E}_{1/3}$ yields $5\mid s_{\text{prim}}$ while every rational point on $\mathcal{E}_{3/5}$ is 5-integral and yields $5\nmid s_{\text{prim}}$). Consequently, both the $5\mid s$ and $5\nmid s$ sectors contain infinitely many primitive solutions in $(\mathbb{Z}\setminus\{0\})^5$. Finally, we analyze the Three-Squares Identity, the Four-Squares Surface S_0 ($2a+b+c=0$) and its genus-3 and genus-1 pencils, the 3-parameter conic-bundle family in (e_0,m,t), the 6-point and 5-point osculating conic maps, the rational elimination of d, the Centered Double-Depressed-Cubic Canonical Form, the Paired-Solution Surface ($u=2e$) and its genus-3 fibers over genus-1 Weierstrass quotients, the Opposite-Pair ($e=-d$) and Equal-Pair ($d=e$) surfaces, and the structural classification of the 42-row census.

Contents

Empirical Census of Primitive Solutions ($0<a+b+c+d+e<1000$) **3**

1 Explicit Elliptic-Conic Parametrization ($e_0=-1$ Sector) **4**

2 Strict Convexity Obstruction, Norm-Form Identities, and Bitangent Geometry **7**

2.1 Smoothness of $V(\mathbb{C})$ and Strict Convexity of $V(\mathbb{R})$ 7

2.2 The Three-Squares and Four-Squares Identities 8

2.3 Bitangent Hyperplanes via Complex Conjugate Lines on $V(\mathbb{C})$ 9

3 Number-Theoretic Structure of the Solutions Table (mod 5 and mod 2 Dichotomies) 10

4 General 3-Parameter Conic-Bundle Reduction in (e_0, m, t) 10

5 Universal Bitangent-Plane Parametrization and the $e_0 = 2$ Sector 11

6 General 6-Point Osculating-Parabola Map Around Symmetric Solutions ($b_0 = c_0$) 13

7 Rational Elimination of d and the Separation-of-Variables Surface 14

8 Universal Tangent-Slope Shift, Integral Weierstrass Models, and Closed-Form Rational Recurrence 15

9 Explicit Parametrization for the $5 \nmid (a+b+c+d+e)$ Sector ($e_0 = \frac{3}{5}$) 16

10 General Non-Even Osculating Conic Around an Arbitrary Solution ($b_0 \neq c_0$) 18

11 Depressed-Cubic Reduction on the Separated Surface for Arbitrary $(e, p) \in \mathbb{Q}^2$ 19

12 The Centered Double-Depressed-Cubic Identity (Global Canonical Form) 20

13 The Paired-Solution Surface ($u = 2e$) and Weierstrass Fibrations 21

14 Explicit Pencils on the Four-Squares Surface S_0 ($2a + b + c = 0$) 21

15 The Opposite-Pair Surface ($e = -d$) and Its Sum-of-Fourth-Powers Identity 22

16 The Equal-Pair Surface ($d = e$) and Dual Osculating Families 22

17 The Hyperbolic $(s-d)(s-e)$ Factorization on the Three-Squares Identity 23

18 Explicit Polynomial Parametrization for the $e_0 = \frac{1}{3}$ Sector (Row 7) 23

19 Secant-Interpolation Cubic and Slope Map on the Paired-Solution Surface ($u = 2e$) 25

20 Classification of Algebraic Subfamilies and Complete 42-Row Census Index 26

A Reproducible C99 Verification Code for the $0 < s < 1000$ Census 29

Empirical Census of Primitive Solutions ($0 < a+b+c+d+e < 1000$)

Before developing the parametric theory in Sections 1–20, we establish the finite compactness bound, local 2-adic parity law, and two-sum fourth-power monotonicity that make an exhaustive enumeration over $0 < s = a+b+c+d+e < 1000$ finite and independently reproducible, and we record the complete table of 42 primitive integer solutions in $(\mathbb{Z} \setminus \{0\})^5$. Complete C99 source code for two independent verification algorithms and the execution log are provided in Appendix A.

Proposition 0.1 (Compactness Bound, 2-Adic Parity Law, and Exhaustiveness of the Census)**.** *Let $(a,b,c,d,e) \in (\mathbb{Z} \setminus \{0\})^5$ be a primitive integer solution to $a^4+b^4+c^4+d^4+e^4 = s^4$ with $s = a+b+c+d+e > 0$ and $\gcd(a,b,c,d,e) = 1$. Then:*

1. ***Compactness Bound (Absence of Unbounded Cancellation):*** *Since $x^4 \geq 1$ for every $x \in \mathbb{Z} \setminus \{0\}$, each coordinate satisfies $x_i^4 = s^4 - \sum_{j \neq i} x_j^4 \leq s^4 - 4$, and therefore*

$$1 \leq |a|, |b|, |c|, |d|, |e| \leq \left\lfloor (s^4-4)^{1/4} \right\rfloor \leq s-1. \tag{1}$$

   In particular, for any upper bound $s < S_{\max}$, the set of solutions in $(\mathbb{Z} \setminus \{0\})^5$ is strictly contained in the finite box $[-S_{\max}+1, S_{\max}-1]^5$.

2. ***2-Adic Parity Law:*** *Modulo 16, every integer satisfies $x^4 \equiv 0 \pmod{16}$ if $2 \mid x$ and $x^4 \equiv 1 \pmod{16}$ if $2 \nmid x$. Hence the number $k \in \{0,1,2,3,4,5\}$ of odd coordinates among (a,b,c,d,e) satisfies $k \equiv s^4 \pmod{16}$. If $2 \mid s$, then $s^4 \equiv 0 \pmod{16}$ forces $k = 0$, so all five coordinates are even, contradicting $\gcd(a,b,c,d,e) = 1$. Thus s is necessarily odd ($s \equiv 1 \pmod 2$), and $s^4 \equiv 1 \pmod{16}$ forces $k = 1$: **exactly one coordinate x_{odd} is odd, while the remaining four coordinates are even** ($2B, 2C, 2D, 2E$ with $B, C, D, E \in \mathbb{Z} \setminus \{0\}$).*

3. ***Meet-in-the-Middle Completeness Verification for $0 < s < 1000$:*** *Writing the four even coordinates in non-increasing order as $2B \geq 2C \geq 2D \geq 2E$ with $1 \leq |B|, |C|, |D|, |E| \leq \frac{s-1}{2} \leq 499$, the Diophantine system for each odd $s \in \{1, 3, \ldots, 999\}$ and each odd $x_{\text{odd}} \in [-s+2, s-2]$ is equivalent to*

$$(B+C) + (D+E) = \frac{s - x_{\text{odd}}}{2}, \qquad (B^4+C^4) + (D^4+E^4) = \frac{s^4 - x_{\text{odd}}^4}{16}. \tag{2}$$

   Moreover, within each two-sum bucket $S_2 = B + C \in [-998, 998]$, the exact algebraic identity

$$8(B^4+C^4) + 8S_2^4 = \left((B-C)^2 + 3S_2^2\right)^2$$

   *shows that $P_2 = B^4 + C^4$ is a strictly increasing function of $B - C \geq 0$, so all $\binom{998+1}{2} = 498{,}501$ ordered non-zero pairs (B, C) ($-499 \leq C \leq B \leq 499$) have strictly distinct (S_2, P_2) keys (0 duplicate P_2 values, with maximum bucket size 499 of 500). Sorting each bucket by P_2, an exact two-pointer scan over (2) across all 249,500 odd pairs (s, x_{odd}) (performing 124,750,000 loop iterations over $(s, x_{\text{odd}}, S_{BC})$, of which 124,749,997 scan non-empty bucket pairs; see Listing 1 in Appendix A), independently confirmed by an unrestricted fourth-power-budget search over all $1 \leq s \leq 999$ including even s (Listing 3, finding 408 total non-zero solutions, none with even s), exhaustively enumerates all primitive solutions with $0 < s < 1000$, yielding **36 primitive solutions with $5 \mid s$** (Table I) and **6 primitive solutions with $5 \nmid s$** (Table II) up to permutation of (a,b,c,d,e). Rows 1–14 of Table I are precisely the primitive solutions with $\max_i |x_i| \leq 300$.*

Table I: All Primitive Solutions with $a+b+c+d+e$ Divisible by 5 for $0 < s < 1000$ (36 Solutions)

Row	s	a	b	c	d	e	**Row**	s	a	b	c	d	e
1	5	−3	4	4	2	−2	19	585	97	514	432	−332	−126
2	15	13	12	−6	−2	−2	20	625	497	488	−432	76	−4
3	15	−9	14	8	6	−4	21	625	619	−268	156	62	56
4	135	111	−102	92	26	8	22	645	317	582	−468	116	98
5	135	111	114	−58	−16	−16	23	645	637	−276	226	64	−6
6	175	173	−72	62	26	−14	24	685	37	664	382	−252	−146
7	205	−121	182	146	−18	16	25	705	−447	644	432	128	−52
8	225	57	186	184	−104	−98	26	705	667	−416	372	54	28
9	235	73	216	−156	126	−24	27	715	681	392	−388	28	2
10	295	143	−262	216	124	74	28	755	289	748	−262	−108	88
11	295	277	−198	104	74	38	29	775	−657	604	418	322	88
12	315	−253	264	168	98	38	30	835	−117	−674	628	502	496
13	325	111	294	−242	96	66	31	835	587	668	−572	474	−322
14	355	−271	298	224	96	8	32	885	429	−808	554	488	222
15	425	−63	344	312	−306	138	33	905	−583	834	514	202	−62
16	435	123	368	−354	194	104	34	945	811	724	−546	−176	132
17	515	203	458	−394	144	104	35	965	−697	784	706	264	−92
18	515	−407	352	346	338	−114	36	985	291	936	576	−426	−392

Table II: All Primitive Solutions with $a+b+c+d+e$ Not Divisible by 5 for $0<s<1000$ (6 Solutions)

Row	s	a	b	c	d	e	$s-a$
1 (37)	363	−262	315	230	50	30	$625=5^4$
2 (38)	413	−212	55	390	250	−70	$625=5^4$
3 (39)	507	−118	395	450	−150	−70	$625=5^4$
4 (40)	551	−74	335	490	−380	180	$625=5^4$
5 (41)	967	342	−465	900	600	−410	$625=5^4$
6 (42)	991	−259	930	−590	500	410	$1250=2\cdot 5^4$

1 Explicit Elliptic-Conic Parametrization ($e_0=-1$ Sector)

We begin by presenting a complete, self-contained quadratic polynomial parametrization over a smooth genus-1 curve that provably generates infinitely many primitive non-zero integer solutions to $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$ in the $5\mid s$ sector.

Theorem 1.1 (Explicit Parametrization and Infinitude in the $e_0=-1$ Bitangent Sector)**.** *Define the five polynomials (a,b,c,d,e) in the rational variables (u,v) by*

$$\begin{aligned} a(u,v) &= 8u^2-76u-432,\\ b(u,v) &= -5u^2+38u+270+v,\\ c(u,v) &= -5u^2+38u+270-v,\\ d(u,v) &= 4u^2-216,\\ e(u,v) &= 8u^2+76u-432, \end{aligned} \tag{3}$$

whose sum is

$$s(u,v)=a+b+c+d+e=10u^2+76u-540. \tag{4}$$

Then:

1. ***Algebraic Identity:*** *The fourth-power relation $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$ holds identically whenever $(u,v)\in\mathbb{Q}^2$ lies on the smooth genus-1 quartic curve*

$$\mathcal{E}: \quad v^2=f(u)=-151u^4+1140u^3+3768u^2-61560u+2916. \tag{5}$$

2. ***Integral Weierstrass Model and Infinite Order*** *(*$\operatorname{rank} E(\mathbb{Q}) \geq 1$*):* *Taking $Q_0=(0,-54)\in\mathcal{E}(\mathbb{Q})$ as the identity element $\mathcal{O}$, the birational transformation*

$$X_W=\frac{432(54-570u-v)}{u^2},\qquad Y_W=432\left[2\left(\frac{X_W^2}{186624}+151\right)u+\frac{95}{36}X_W-1140\right] \tag{6}$$

maps $\mathcal{E}$ isomorphically onto the nonsingular integral Weierstrass cubic

$$E:\quad Y_W^2=X_W^3+15072X_W^2-1094674176X_W-35955750223872. \tag{7}$$

Under (6), the base point $P_0=(0,54)\in\mathcal{E}(\mathbb{Q})$ maps to the integral point $P_0^W=(1284528,\ 1463869440)\in E(\mathbb{Z})$, whereas its double $2P_0^W$ has non-integral X_W-coordinate

$$X_W(2P_0^W)=\frac{61581496441}{193600}=\frac{61581496441}{440^2}\notin\mathbb{Z}.$$

By the Nagell–Lutz Theorem, P_0^W has ***infinite order*** *in $E(\mathbb{Q})$ (so $\operatorname{rank} E(\mathbb{Q})\geq 1$).*

3. ***Generic Injectivity, Zero-Coordinate Avoidance, and Universal 5-Adic Sector Invariance:*** *The rational map $\Phi:\mathcal{E}\dashrightarrow V\subset\mathbb{P}^4$ defined by (3) maps $\mathcal{E}$ birationally (with degree 1, i.e., generically injectively) onto its image curve inside the 2-plane $2(a+b+c)+d=a+e-4d=0$. Every rational point $(u,v)\in\mathcal{E}(\mathbb{Q})$ outside at most 24 points where $abcdes=0$ yields, after clearing denominators and dividing by the greatest common divisor, a primitive integer solution in $(\mathbb{Z}\setminus\{0\})^5$ satisfying $5\mid s_{\text{prim}}$. Consequently, $\langle P_0^W\rangle\subset E(\mathbb{Q})$ generates infinitely many distinct primitive integer solutions with $5\mid s$.*

Proof. Set $x=\frac{5u^2-38u-270}{2(u^2-54)}$ and $y=\frac{v}{2(u^2-54)}$ (for $u^2\neq 54$). Clearing the common denominator $2(u^2-54)$ from the affine linear system

$$a=-2x+1,\quad b=x+y,\quad c=x-y,\quad d=-2,\quad e=2x-9,\quad s=2x-10$$

(or equivalently multiplying the system $a=2x-1$, $b=-x+y$, $c=-x-y$, $d=2$, $e=-2x+9$, $s=-2x+10$ by $2(u^2-54)$ with $y\mapsto -y$) produces the polynomials in (3). Direct expansion of $\frac{1}{2}[s^4-(a^4+b^4+c^4+d^4+e^4)]$ in (x,y) yields

$$\begin{aligned}\frac{s^4-(a^4+b^4+c^4+d^4+e^4)}{2}&=\frac{(-2x+10)^4-(2x-1)^4-2^4-(-2x+9)^4}{2}-\frac{(-x+y)^4+(-x-y)^4}{2}\\&=(-9x^4+216x^2-1080x+1711)-(x^4+6x^2y^2+y^4)\\&=(216x^2-1080x+1711)-(3x^2+y^2)^2.\end{aligned}$$

Substituting $x=\frac{5u^2-38u-270}{2(u^2-54)}$ gives

$$216x^2-1080x+1711=\left(\frac{19(u^2+54)}{u^2-54}\right)^2,$$

while $3x^2+y^2=\frac{3(5u^2-38u-270)^2+v^2}{4(u^2-54)^2}$. Imposing $3x^2+y^2=-\frac{19(u^2+54)}{u^2-54}$ and clearing the denominator $4(u^2-54)^2$ gives

$$3(5u^2-38u-270)^2+v^2=-76(u^2-54)(u^2+54)=-76(u^4-2916),$$

which simplifies directly to $v^2=-151u^4+1140u^3+3768u^2-61560u+2916$. The quartic polynomial $f(u)$ has non-zero discriminant

$$\Delta(f)=-2^{18}\cdot 3^{10}\cdot 19^5\cdot 229\cdot 506689=-4447310299766465398308864\neq 0,$$

so $\mathcal{E}$ is smooth of genus 1.

For part (2), substituting $v = 54 - 570u - \frac{X_W}{432}u^2$ into $v^2 - f(u) = 0$ cancels u^0 and u^1; dividing by u^2 leaves the quadratic equation in u

$$\left(\frac{X_W^2}{186624} + 151\right)u^2 + \left(\frac{95}{36}X_W - 1140\right)u + \left(321132 - \frac{X_W}{4}\right) = 0,$$

whose discriminant with respect to u, multiplied by $432^2 = 186624$, equals $Y_W^2 = X_W^3 + 15072X_W^2 - 1094674176X_W - 35955750223872$, yielding (6)–(7) with inverse map

$$u = \frac{216(Y_W - 1140X_W + 492480)}{X_W^2 + 28180224}, \qquad v = 54 - 570u - \frac{X_W}{432}u^2.$$

Along the branch $v = 54 - 570u - \frac{26761}{9}u^2 + O(u^3)$ at $P_0 = (0, 54)$, we obtain $P_0^W = (1284528, 1463869440) \in E(\mathbb{Z})$. The tangent line to E at P_0^W has slope

$$\lambda = \frac{3X_W(P_0^W)^2 + 30144X_W(P_0^W) - 1094674176}{2Y_W(P_0^W)} = \frac{749579}{440},$$

so the chord-and-tangent duplication formula gives

$$X_W(2P_0^W) = \lambda^2 - 15072 - 2X_W(P_0^W) = \frac{61581496441}{193600} = \frac{61581496441}{440^2} \notin \mathbb{Z}.$$

By the Nagell–Lutz Integrality Theorem for integral Weierstrass models $Y^2 = X^3 + a_2X^2 + a_4X + a_6$ with $a_2, a_4, a_6 \in \mathbb{Z}$ [5, Ch. VIII, Cor. 7.2] (and noting that $\gcd(440, 6) = 2 > 1$, $11 \mid 440$, so $X_W(2P_0^W) + \frac{a_2}{3} = X_W(2P_0^W) + 5024$ also has non-integral denominator 440^2 on the depressed model), every non-zero rational torsion point on $E(\mathbb{Q})$ has integer coordinates. Because $X_W(2P_0^W) \notin \mathbb{Z}$, $2P_0^W$ is not torsion, so P_0^W has infinite order.

For part (3), the image of Φ satisfies both $2(a + b + c) + d = 0$ and $a + e = 4d$. From a point $[a : b : c : d : e]$ in the image with $d \neq 0$, the ratios

$$\frac{e(u) - a(u)}{d(u)} = \frac{38u}{u^2 - 54}, \qquad \frac{b(u, v) - c(u, v)}{d(u)} = \frac{v}{2(u^2 - 54)}$$

determine (u, v) up to the algebraic involution $(u, v) \mapsto (-54/u, -54v/u^2)$. However, $(u', v') = (-54/u, -54v/u^2)$ does not lie on $\mathcal{E}$ dynamically, since $u^4 f(-54/u) - 54^2 f(u) = 443232(u^4 - 2916) \not\equiv 0$; equivalently, since $\frac{3(b+c)^2+(b-c)^2}{d^2} = \frac{19(2916-u^4)}{(u^2-54)^2}$ on $\mathcal{E}$, u and v are recovered uniquely as rational functions on the image curve via

$$u = \frac{-54\,\frac{e-a}{d}}{\frac{3(b+c)^2+(b-c)^2}{d^2} + 19}, \qquad v = 2(u^2 - 54)\frac{b - c}{d}.$$

Thus Φ has degree 1 onto its image curve, and at most $5! = 120$ points of $\mathcal{E}(\mathbb{Q})$ can projectively represent the same primitive solution up to permutation. None of the six functions a, b, c, d, e, s vanishes identically on $\mathcal{E}$ (all six are non-zero at P_0), and each has a pole divisor of degree 4 (order 2 at each of the two points at infinity on the smooth projective model of $\mathcal{E}$), so each has at most 4 zeros on $\mathcal{E}$ (in particular, $b(u, v)c(u, v) = (-5u^2 + 38u + 270)^2 - f(u) = 0$ has 4 roots in u, each giving one point with $b = 0$ and one with $c = 0$). Thus $abcdes = 0$ defines at most $6 \times 4 = 24$ points on $\mathcal{E}(\mathbb{Q})$. Finally, for any rational point $(u, v) = (p/q, V/q^2) \in \mathcal{E}(\mathbb{Q})$ with $\gcd(p, q) = 1$:

- If $5 \nmid q$, then $f(u) \bmod 5$ takes the values $(1, 3, 2, 2, 3)$ for $u \equiv (0, 1, 2, 3, 4) \pmod 5$. Since 2 and 3 are quadratic non-residues modulo 5, $v^2 = f(u)$ forces $u \equiv 0 \pmod 5$, at which $s(u) = 10u^2 + 76u - 540 \equiv 0 \pmod 5$ while $d(u) = 4u^2 - 216 \equiv 4 \not\equiv 0 \pmod 5$.
- If $5 \mid q$ (so $5 \nmid p$), then $q^2 s(p/q) = 10p^2 + 76pq - 540q^2 \equiv 0 \pmod 5$ while $q^2 d(p/q) = 4p^2 - 216q^2 \equiv 4p^2 \not\equiv 0 \pmod 5$.

In both cases, $v_5(q^2 s(p/q)) \geq 1 > 0 = v_5(q^2 d(p/q))$, so removing $\gcd(q^2 a, \ldots, q^2 e)$ preserves $5 \mid s_{\text{prim}}$ for every rational point on $\mathcal{E}(\mathbb{Q})$. □

Explicit Generation of Primitive Solutions from Theorem 1.1

1. **Base Point ($u=0, v=54$):** At $(u,v)=(0,54)$, (3) gives the tuple $(-432, 324, 216, -216, -432)$. Dividing by -108 yields Row 1 of Table I (up to permutation):

$$(a,b,c,d,e)=(4,-3,-2,2,4), \qquad s=5.$$

2. **First Osculating Tangent Step ($2P_0^W$ on E):** Expanding $v=\sqrt{f(u)}$ at $(0,54)$ to second order in u defines the osculating parabola

$$v(u)=54-570u-\frac{26761}{9}u^2.$$

Then $v(u)^2-(-151u^4+1140u^3+3768u^2-61560u+2916)=\frac{2888}{81}u^3(247979u+95040)=0$. The non-zero root and its corresponding ordinate on $\mathcal{E}$ are

$$u=-\frac{95040}{247979}, \qquad v=-\frac{10103568715386}{61493584441}.$$

Substituting (u,v) into (3), clearing the denominator $247979^2=61493584441$, and dividing by the greatest common divisor $-2052=-19\cdot 108$ (where $2052=2^2\cdot 3^3\cdot 19$, with $b\leftrightarrow c$ corresponding to $\pm v$) yields the exact 11-digit primitive solution:

$$\begin{aligned} a&=12037916876, & b&=-2709042377, & c&=-12556575238,\\ d&=6455401478, & e&=13783689036, & s&=17011389775. \end{aligned}$$

2 Strict Convexity Obstruction, Norm-Form Identities, and Bitangent Geometry

In this section we prove that $V(\mathbb{C})$ is smooth, explain why no real straight line exists on $V(\mathbb{R})$, establish the Three-Squares and Four-Squares Identities, and derive the rational bitangent hyperplane family H_{e_0} and its conic-bundle structure.

2.1 Smoothness of $V(\mathbb{C})$ and Strict Convexity of $V(\mathbb{R})$

Proposition 2.1 (Smoothness of $V(\mathbb{C})$, Strict Convexity, and Absence of Real Lines)**.** *Let $V\subset\mathbb{P}^4$ be the projective quartic threefold defined by $F(x_1,x_2,x_3,x_4,x_5)=\sum_{i=1}^5 x_i^4-(\sum_{i=1}^5 x_i)^4=0$. Then:*

1. ***Global Smoothness over $\mathbb{C}$:*** *The threefold $V(\mathbb{C})$ is nonsingular everywhere in $\mathbb{P}^4(\mathbb{C})$.*

2. ***Strict Convexity and Absence of Real Lines:*** *In the affine chart $s=a+b+c+d+e=1$, the real locus $V(\mathbb{R})$ is defined by $g(a,b,c,d)=a^4+b^4+c^4+d^4+(1-a-b-c-d)^4=1$. The function $g:\mathbb{R}^4\to\mathbb{R}$ is globally strictly convex on $\mathbb{R}^4$, and its Hessian matrix D^2g is strictly positive definite at every point of $V(\mathbb{R})$ having at most one zero coordinate. Consequently, $V(\mathbb{R})$ bounds a strictly convex compact body in $\mathbb{R}^4$ and **contains no real straight lines**, and for any real point $P\in V(\mathbb{R})$ with non-zero coordinates, the real tangent hyperplane $T_PV(\mathbb{R})$ meets $V(\mathbb{R})$ solely at P.*

3. ***Real Plane Sections:*** *Any real 2-plane $\Pi\subset\mathbb{P}^4(\mathbb{R})$ intersecting $V(\mathbb{R})$ in more than one real point and avoiding the coordinate-zero locus can only be tangent to $V(\mathbb{C})$ at non-real complex conjugate pairs of points $P,\overline{P}\in V(\mathbb{C})$. Thus the plane quartic $C=V\cap\Pi$ has no real singular points and its total delta-invariant $\delta=\sum\delta_Q$ is even: if C is irreducible over $\mathbb{C}$, then $\delta\le 3$ forces $\delta\in\{0,2\}$ (geometric genus $g=3-\delta\in\{3,1\}$), whereas if C is reducible without real singularities, it splits into a complex conjugate pair of conics meeting in 4 non-real points ($\delta=4$).*

Proof. For part (1), a singular point $[x_1:\cdots:x_5]\in V(\mathbb{C})$ must satisfy $\frac{\partial F}{\partial x_i}=4(x_i^3-s^3)=0$ for all $i=1,\dots,5$, where $s=\sum_{i=1}^5 x_i$. If $s=0$, then $x_i^3=0$ for all i, which is impossible in $\mathbb{P}^4(\mathbb{C})$. If $s\neq 0$, scaling so that $s=1$ forces $x_i^3=1$, so $x_i\in\{1,\omega,\omega^2\}$ for each $i=1,\dots,5$. If $k\in\{0,\dots,5\}$ of the coordinates equal 1 and $5-k$ belong to $\{\omega,\omega^2\}$, taking real parts of $\sum_{i=1}^5 x_i=1$ gives $k-\frac{5-k}{2}=\frac{3k-5}{2}=1$, which has no integer solution $k\in\mathbb{Z}$. Hence $V(\mathbb{C})$ is smooth.

For parts (2) and (3), for any distinct points $x\neq y\in\mathbb{R}^4$ and $\lambda\in(0,1)$, setting $x_5=1-\sum_{i=1}^4 x_i$ and $y_5=1-\sum_{i=1}^4 y_i$ implies $x_i\neq y_i$ for at least one $i\in\{1,\dots,5\}$. Because $t\mapsto t^4$ is strictly convex on $\mathbb{R}$, we have $g((1-\lambda)x+\lambda y)<(1-\lambda)g(x)+\lambda g(y)$, so g is globally strictly convex on $\mathbb{R}^4$ and the level set $g=1$ contains no real line segments or real straight lines. Computing the second partial derivatives gives $\frac{\partial^2 g}{\partial x_i\partial x_j}=12x_i^2\delta_{ij}+12x_5^2$, so for any non-zero vector $v=(v_1,v_2,v_3,v_4)^T\in\mathbb{R}^4$,

$$v^T(D^2g)v=12\left(\sum_{i=1}^4 x_i^2v_i^2+x_5^2\left(\sum_{i=1}^4 v_i\right)^2\right)>0$$

whenever at most one of $x_1,\dots,x_5$ vanishes. Equivalently, in homogeneous coordinates, at any real point $P=(x_1,\dots,x_5)\in V(\mathbb{R})$ with non-zero coordinates, a tangent direction v satisfies $\sum_{i=1}^5 x_i^3v_i=s^3v_s$ (with $v_s=\sum_{i=1}^5 v_i$), and by the Cauchy–Schwarz inequality,

$$(s^3v_s)^2=\left(\sum_{i=1}^5(x_i^2)(x_iv_i)\right)^2\le\left(\sum_{i=1}^5 x_i^4\right)\left(\sum_{i=1}^5 x_i^2v_i^2\right)=s^4\sum_{i=1}^5 x_i^2v_i^2,$$

with equality if and only if v is proportional to P (which is $v=0$ in $T_P\mathbb{P}^4$). Thus the second fundamental form is strictly definite at all non-zero-coordinate points. Consequently, if a real 2-plane Π avoiding the coordinate-zero locus meets $V(\mathbb{R})$ in more than one real point, $C=V\cap\Pi$ has no real singular points; its singularities occur strictly in non-real complex conjugate pairs, so $\delta=\sum\delta_Q$ is even. For an irreducible plane quartic, the arithmetic genus is $p_a=3$, so $g=3-\delta\ge 0$ forces $\delta\in\{0,2\}$ ($g\in\{3,1\}$); if C is reducible over $\mathbb{C}$ with no real singularities, it cannot have a real line component, hence must be a conjugate pair of conics ($\delta=4$). □

2.2 The Three-Squares and Four-Squares Identities

Theorem 2.2 (Three-Squares Identity and Four-Squares Surface S_0)**.** *Write $b=-x+y$, $c=-x-y$, $d=-z+w$, and $e=-z-w$ (so $x=-\frac{b+c}{2}$, $y=\frac{b-c}{2}$, $z=-\frac{d+e}{2}$, $w=\frac{d-e}{2}$, and $s=a-2x-2z$). Then the fourth-power equation $a^4+b^4+c^4+d^4+e^4=s^4$ is **identically equivalent** to the Three-Squares Identity:*

$$\begin{aligned}\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2-\left(4(x^2+xz+z^2)-2a(x+z)+a^2\right)^2\\ =-a\left(a^3-8axz+16xz(x+z)\right).\end{aligned}\tag{8}$$

*In particular, on the hyperplane $2a+b+c=0$ (i.e., $a=x$), the cubic remainder factors as $-x^2(x+4z)^2$, reducing the equation to the **Four-Squares Surface** S_0:*

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2+x^2(x+4z)^2=\left(3x^2+2xz+4z^2\right)^2,\tag{9}$$

or equivalently, in factored norm form:

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2=4(x^2-xz+2z^2)(2x^2+3xz+2z^2).\tag{10}$$

Proof. Since $(-x+y)^4+(-x-y)^4=2x^4+12x^2y^2+2y^4=2(y^2+3x^2)^2-16x^4$ and similarly $d^4+e^4=2(w^2+3z^2)^2-16z^4$, the equation $a^4+b^4+c^4+d^4+e^4-(a-2x-2z)^4=0$ divides by 2 to give

$$(y^2+3x^2)^2+(w^2+3z^2)^2=\frac{(a-2x-2z)^4-a^4}{2}+8x^4+8z^4.$$

Expanding $\left(4(x^2+xz+z^2)-2a(x+z)+a^2\right)^2$ and subtracting from the right-hand side yields

$$\begin{aligned}&\left[-4a^3(x+z)+12a^2(x+z)^2-16a(x+z)^3+16(x^2+xz+z^2)^2\right]\\&\quad-\left[4(x^2+xz+z^2)-2a(x+z)+a^2\right]^2\\&\quad=-a^4+8a^2xz-16axz(x+z)=-a(a^3-8axz+16xz(x+z)),\end{aligned}$$

proving (8). Setting $a=x$ gives $a^3-8axz+16xz(x+z)=x(x^2+8xz+16z^2)=x(x+4z)^2$ and $4(x^2+xz+z^2)-2x(x+z)+x^2=3x^2+2xz+4z^2$, establishing (9) and (10). □

Remark 2.3 (Symmetric Newton Identity Form)**.** In terms of $p=d+e$, $q=de$, $\sigma_1=a+b+c$, $\sigma_2=ab+bc+ca$, and $\sigma_3=abc$, Newton's identities yield the polynomial identity

$$\frac{(a+b+c+d+e)^4-(a^4+b^4+c^4+d^4+e^4)}{2}=(p^2+p\sigma_1+\sigma_1^2)^2-(d^2+de+e^2)^2-(\sigma_1^2-\sigma_2)^2-2\sigma_1\sigma_3.$$

Similarly, grouping four variables into two pairs $p_1=a+b$ and $p_2=c+d$, the Diophantine equation $(a+b+c+d+e)^4=a^4+b^4+c^4+d^4+e^4$ is equivalent to the relation

$$(a^2+ab+b^2)^2+(c^2+cd+d^2)^2-\left(p_1^2+p_1p_2+p_2^2+e(p_1+p_2)\right)^2=2e(e+p_1)(e+p_2)(p_1+p_2).$$

2.3 Bitangent Hyperplanes via Complex Conjugate Lines on $V(\mathbb{C})$

Theorem 2.4 (Rational Bitangent Hyperplane Family H_{e_0})**.** *Let $\omega=\frac{-1+i\sqrt{3}}{2}$ be a primitive cube root of unity. Then the complex conjugate projective lines*

$$\begin{aligned}L&=\left\{(t:\omega t:\omega^2 t:0:s)\in\mathbb{P}^4(\mathbb{C})\;\middle|\;(t:s)\in\mathbb{P}^1(\mathbb{C})\right\},\\\overline{L}&=\left\{(t:\omega^2 t:\omega t:0:s)\in\mathbb{P}^4(\mathbb{C})\;\middle|\;(t:s)\in\mathbb{P}^1(\mathbb{C})\right\}\end{aligned}$$

lie entirely on $V(\mathbb{C})$ inside the base 2-plane $\Pi_0=\{a+b+c=0,\,d=0\}$, along which $V\cap\Pi_0=2L+2\overline{L}$ (since $a^4+b^4+(-a-b)^4+0^4+e^4-(0+e)^4=2(a^2+ab+b^2)^2=2(a-\omega b)^2(a-\omega^2 b)^2$). For any $e_0\in\mathbb{Q}\setminus\{0,1\}$, the common tangent hyperplane to $V(\mathbb{C})$ at $P_\zeta=(1:\omega:\omega^2:0:\zeta e_0)\in L$ and $\overline{P}_\zeta=(1:\omega^2:\omega:0:\zeta e_0)\in\overline{L}$ depends only on e_0^3, coincides for all three cube roots of unity $\zeta\in\{1,\omega,\omega^2\}$, is defined over $\mathbb{Q}$, and is tangent to $V(\mathbb{C})$ at all ***six points*** *$\{P_1,P_\omega,P_{\omega^2},\overline{P}_1,\overline{P}_\omega,\overline{P}_{\omega^2}\}$:*

$$H_{e_0}:\quad(e_0^3-1)(a+b+c)+e_0^3d=0\iff\frac{a+b+c}{a+b+c+d}=e_0^3\quad(a+b+c+d\neq0).\tag{11}$$

Every hyperplane H_{e_0} contains the base plane Π_0, and the quartic surface $S_{e_0}=V\cap H_{e_0}$ is singular at the six tangency points. Parametrizing H_{e_0} in coordinates $(x:y:z:w)\in\mathbb{P}^3$ by

$$a=2x+e_0^3z,\quad b=-x+y,\quad c=-x-y,\quad d=(1-e_0^3)z,\quad e=2e_0x+w\quad(s=z+e)\tag{12}$$

cancels both the x^4 and x^3 terms of $s^4-(a^4+b^4+c^4+d^4+e^4)=0$, reducing $S_{e_0}=V\cap H_{e_0}$ to a quadratic conic bundle in x:

$$\left(3x^2+y^2\right)^2=z\left[12e_0^2\left(2w+(1-e_0^4)z\right)x^2+4e_0\left(3w^2+3wz+(1-e_0^8)z^2\right)x+C_{e_0}(w,z)\right],\tag{13}$$

where $C_{e_0}(w,z)=2w^3+3w^2z+2wz^2-e_0^3(e_0^9-2e_0^6+3e_0^3-2)z^3$.

Proof. Along L, we have $a+b+c+d+e=(1+\omega+\omega^2)t+0+s=s$ and $a^4+b^4+c^4+d^4+e^4=(1+\omega+\omega^2)t^4+0+s^4=s^4$ identically for all $(t:s)\in\mathbb{P}^1(\mathbb{C})$. At $P_\zeta=(1,\omega,\omega^2,0,\zeta e_0)$ with $\zeta^3=1$, we have $s=\zeta e_0$ and $a^3=b^3=c^3=1$, $d^3=0$, $e^3=\zeta^3e_0^3=e_0^3=s^3$. Since $\frac{\partial F}{\partial x_k}=4(s^3-x_k^3)$, the gradient at all six points $P_\zeta,\overline{P}_\zeta$ ($\zeta\in\{1,\omega,\omega^2\}$) is

$$\nabla F(P_\zeta)=\nabla F(\overline{P}_\zeta)=4(e_0^3-1,e_0^3-1,e_0^3-1,e_0^3,0),$$

yielding (11) and showing that $S_{e_0}=V\cap H_{e_0}$ is singular at all six points. Under (12), $b^4+c^4=2(3x^2+y^2)^2-16x^4$, while $a^4=(2x+e_0^3z)^4=16x^4+32e_0^3x^3z+\dots$ cancels $-16x^4$, and $s^4-e^4=(z+2e_0x+w)^4-(2e_0x+w)^4=4z(2e_0x+w)^3+\dots$ has leading x^3 term $32e_0^3x^3z$, which cancels the $32e_0^3x^3z$ term from a^4, leaving (13). □

3 Number-Theoretic Structure of the Solutions Table (mod 5 and mod 2 Dichotomies)

Theorem 3.1 (5-Adic Divisibility and Modulo-10 Structure Law)**.** *Let $(a,b,c,d,e)\in(\mathbb{Z}\setminus\{0\})^5$ be a primitive solution to $a^4+b^4+c^4+d^4+e^4=s^4$ with $s=a+b+c+d+e$. By Proposition 0.1, s is odd and exactly one coordinate x_{odd} is odd while four coordinates are even. Modulo 5 and modulo 10:*

1. ***Case 1 ($5\mid s$):*** *None of a,b,c,d,e is divisible by 5: $a^4\equiv b^4\equiv c^4\equiv d^4\equiv e^4\equiv 1 \pmod 5$. Because s is odd and divisible by 5, every primitive solution in this sector satisfies $s\equiv 5 \pmod{10}$ (as seen across all 36 rows of Table I).*

2. ***Case 2 ($5\nmid s$):*** *Exactly four of the variables (say b,c,d,e) are divisible by 5, while the remaining variable satisfies $a\equiv s\not\equiv 0 \pmod 5$ and the strict 5^4-divisibility law:*

$$625=5^4\,\Big|\,(s-a)=b+c+d+e. \tag{14}$$

Furthermore, writing $s-a=625k$ ($k\in\mathbb{Z}\setminus\{0\}$): if k is odd (Rows 1–5 of Table II, $k=1$), then $a=s-625k$ is even, so one of b,c,d,e is congruent to $\pm5 \pmod{10}$ and the other three are divisible by 10; if k is even (Row 6 of Table II, $k=2$), then a is odd and all four of b,c,d,e are divisible by 10 (so $10^4\mid(s^4-a^4)$).

Proof. By Fermat's Little Theorem, $x^4\equiv 0 \pmod 5$ if $5\mid x$ and $x^4\equiv 1 \pmod 5$ if $5\nmid x$. Modulo 5, the sum of five fourth powers is congruent to the number of terms not divisible by 5. If $5\mid s$, then $s^4\equiv 0 \pmod 5$, so either zero or all five of (a,b,c,d,e) are divisible by 5; by primitivity, none is divisible by 5. Since $2\nmid s$ by Proposition 0.1, $s\equiv 5 \pmod{10}$. If $5\nmid s$, then $s^4\equiv 1 \pmod 5$, so exactly one variable (say a) is not divisible by 5, while $5\mid b,c,d,e$. Then $s=a+b+c+d+e\equiv a \pmod 5$, and

$$(s-a)(s+a)(s^2+a^2)=s^4-a^4=b^4+c^4+d^4+e^4\equiv 0 \pmod{5^4}.$$

Since $s\equiv a\not\equiv 0 \pmod 5$, we have $s+a\equiv 2s\not\equiv 0 \pmod 5$ and $s^2+a^2\equiv 2s^2\not\equiv 0 \pmod 5$. Thus $\gcd((s+a)(s^2+a^2),5)=1$, forcing $5^4=625\mid(s-a)$. Combining with the 2-Adic Parity Law of Proposition 0.1(2) yields the modulo-10 classification of b,c,d,e according to the parity of $k=(s-a)/625$. □

4 General 3-Parameter Conic-Bundle Reduction in (e_0,m,t)

We now show how a 1-parameter algebraic substitution in w/z makes the leading x^2z^2 coefficient on H_{e_0} a perfect square without choosing a base point, while noting the real-point obstruction on the resulting quartics.

Theorem 4.1 (General 3-Parameter Conic-Bundle Family)**.** *For any rational parameters $e_0\in\mathbb{Q}\setminus\{0,\pm1\}$ and $m\in\mathbb{Q}$, set $w=\frac{e_0^4-1+3m^2}{2}z$ in Theorem 2.4. Then the coefficient of x^2z^2 on the right-hand side of (13) becomes identically a perfect square, $A_0=(6e_0m)^2$, and the quartic surface S_{e_0} reduces to*

$$\left(3x^2+y^2\right)^2=z^2\left[(6e_0m)^2x^2+B_0(e_0,m)xz+C_0(e_0,m)z^2\right], \tag{15}$$

where

$$B_0(e_0,m)=-e_0\left(e_0^8-18e_0^4m^2-27m^4-1\right), \tag{16}$$

$$C_0(e_0,m)=\frac{27m^6+27e_0^4m^4+3(3e_0^8+1)m^2-3e_0^{12}+8e_0^9-12e_0^6+e_0^4+8e_0^3-2}{4}. \tag{17}$$

Imposing $3x^2+y^2=(6e_0mx+tz)z$ solves linearly for $\frac{x}{z}=\frac{C_0-t^2}{12e_0mt-B_0}$ whenever $12e_0mt-B_0(e_0,m)\neq 0$, giving the 3-parameter polynomial family:

$$\begin{aligned} a(e_0,m;t) &= 2(C_0-t^2)+e_0^3(12e_0mt-B_0),\\ b(e_0,m;t,Y) &= -(C_0-t^2)+Y,\\ c(e_0,m;t,Y) &= -(C_0-t^2)-Y,\\ d(e_0,m;t) &= (1-e_0^3)(12e_0mt-B_0),\\ e(e_0,m;t) &= 2e_0(C_0-t^2)+\frac{e_0^4-1+3m^2}{2}(12e_0mt-B_0), \end{aligned} \tag{18}$$

where (t,Y) lies on the 2-parameter family of quartic curves (smooth of genus 1 whenever $m\neq 0$ and the quartic discriminant in t is non-zero):

$$\mathcal{E}_{e_0,m}: \quad Y^2=(12e_0mt-B_0)\left[6e_0m(C_0-t^2)+t(12e_0mt-B_0)\right]-3(C_0-t^2)^2. \tag{19}$$

Proof. In (13), the coefficient of x^2 (for $z=1$) is $12e_0^2(2w+1-e_0^4)$. Substituting $2w+1-e_0^4=3m^2$ yields $12e_0^2(3m^2)=(6e_0m)^2$. Expanding $(6e_0mx+tz)^2=36e_0^2m^2x^2+12e_0mtxz+t^2z^2$ and equating to $36e_0^2m^2x^2+B_0xz+C_0z^2$ cancels $36e_0^2m^2x^2$ and leaves $(12e_0mt-B_0)x=(C_0-t^2)z$. For $12e_0mt-B_0\neq 0$, substituting into $y^2=(6e_0mx+tz)z-3x^2$ and clearing $(12e_0mt-B_0)^2$ gives $\mathcal{E}_{e_0,m}$. □

Corollary 4.2 (Linear Degeneration at $m=0$)**.** *When $m=0$ (i.e., $w=\frac{e_0^4-1}{2}z$), the x^2 term vanishes completely ($A_0=0$), and S_{e_0} becomes linear in x:*

$$\left(3x^2+y^2\right)^2=z^3\left(B_0(e_0,0)x+C_0(e_0,0)z\right),$$

where $B_0(e_0,0)=-e_0(e_0^8-1)\neq 0$ and $C_0(e_0,0)=-\frac{3e_0^{12}-8e_0^9+12e_0^6-e_0^4-8e_0^3+2}{4}$. Setting $3x^2+y^2=uz^2$ expresses x/z as a quadratic polynomial in u for every $e_0\in\mathbb{Q}\setminus\{0,\pm1\}$:

$$\frac{x}{z}=\frac{u^2-C_0(e_0,0)}{B_0(e_0,0)}, \qquad \left(\frac{y}{z}\right)^2=u-3\left(\frac{u^2-C_0(e_0,0)}{B_0(e_0,0)}\right)^2.$$

Remark 4.3 (Real Obstruction on $\mathcal{E}_{e_0,m}$)**.** The quartic in (19) has leading t^4 coefficient $-3-72e_0^2m^2+144e_0^2m^2=72e_0^2m^2-3$ (and at $m=0$ in Corollary 4.2, leading u^4 coefficient $-3/B_0(e_0,0)^2<0$). Even when the leading coefficient is positive, a given fiber (e_0,m) need not possess real or rational points: for example, at $(e_0,m)=(2,1)$ (or on the $m=0$ specialization where the leading coefficient is -3), the right-hand side is negative or admits no known rational points. Thus Theorem 4.1 provides an algebraic parametrization of the conic bundle over the curve $\mathcal{E}_{e_0,m}$, whereas producing actual rational solutions requires choosing $(e_0,w/z)$ from a known base point as in Theorem 5.1.

5 Universal Bitangent-Plane Parametrization and the $e_0=2$ Sector

Theorem 5.1 (Universal Bitangent-Plane Parametrization from Any Base Point on H_{e_0})**.** *Let $P_0=(a_0,b_0,c_0,d_0,e_0^*)$ be any solution satisfying $\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0}=e_0^3$ for some $e_0\in\mathbb{Q}\setminus\{0,1\}$. Set*

$$z_0=a_0+b_0+c_0+d_0\neq 0, \quad x_0=-\frac{b_0+c_0}{2}, \quad y_0=\frac{b_0-c_0}{2}, \quad w_0=e_0^*-2e_0x_0, \quad U_0=3x_0^2+y_0^2.$$

Fixing $z=z_0$ and $w=w_0$ in (13) defines a conic $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ passing through (x_0,U_0). Parameterizing this conic via $U-U_0=t(x-x_0)$ yields (for $t^2-\mathcal{A}\neq 0$) the universal quadratic polynomial parametrization:

$$\begin{aligned} a(t) &= 2(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})+e_0^3z_0(t^2-\mathcal{A}),\\ b(t,Y) &= -(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})+Y,\\ c(t,Y) &= -(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})-Y,\\ d(t) &= (1-e_0^3)z_0(t^2-\mathcal{A}),\\ e(t) &= 2e_0(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})+w_0(t^2-\mathcal{A}), \end{aligned} \tag{20}$$

where (t, Y) satisfies the quartic curve

$$Y^2=(t^2-\mathcal{A})\left(-U_0t^2+(2\mathcal{A}x_0+\mathcal{B})t-\mathcal{A}U_0\right)-3(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})^2. \tag{21}$$

*Moreover, the resulting rational map $\Phi:\mathcal{E}_{e_0}\dashrightarrow V\subset\mathbb{P}^4$ is **birational (degree 1) onto its image curve**, because $x=-z_0\frac{b+c}{2(a+b+c+d)}$, $y=z_0\frac{b-c}{2(a+b+c+d)}$, and $U=3x^2+y^2$ recover $t=\frac{U-U_0}{x-x_0}$ and $Y=y(t^2-\mathcal{A})$ uniquely as rational functions of $[a:b:c:d:e]$; and when $\mathcal{E}_{e_0}$ is smooth with non-vanishing coordinate functions, $abcde=0$ defines at most $6\times4=24$ points on $\mathcal{E}_{e_0}$.*

Proof. Substituting $U=U_0+t(x-x_0)$ into $U^2-(\mathcal{A}x^2+\mathcal{B}x+\mathcal{C})=0$ and using $U_0^2=\mathcal{A}x_0^2+\mathcal{B}x_0+\mathcal{C}$ gives

$$(x-x_0)\left[(t^2-\mathcal{A})x-(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B})\right]=0.$$

Solving the non-trivial factor for $x(t)$ and $U(t)=U_0+t(x(t)-x_0)$ (for $t^2-\mathcal{A}\neq0$) and clearing the denominator $t^2-\mathcal{A}$ in (12) establishes (20) and (21). Since $a+b+c+d=z_0(t^2-\mathcal{A})$, dividing $-(b+c)/2$ and $(b-c)/2$ by $(a+b+c+d)/z_0$ recovers $x(t)$ and $y(t,Y)$ uniquely, from which $U(t)=3x(t)^2+y(t,Y)^2$, $t=\frac{U(t)-U_0}{x(t)-x_0}$, and $Y=y(t,Y)(t^2-\mathcal{A})$ are recovered uniquely. □

Example 5.2 (Worked Parametrization and Infinite-Order Proof for $e_0=2$ from Row 6: $s=175$)**.** In Row 6 of Table I, $(a_0,b_0,c_0,d_0,e_0^*)=(62,26,-72,-14,173)$ has $a_0+b_0+c_0=16$ and $a_0+b_0+c_0+d_0=2$, so $\frac{16}{2}=8=2^3\implies e_0=2$. Here $z_0=2$, $x_0=23$, $y_0=49$, $w_0=81$, $U_0=3988$, and $(\mathcal{A},\mathcal{B},\mathcal{C})=(12672,306384,2153824)$. Rescaling $t=8u$ in Theorem 5.1 and dividing all coordinates by 16 (with $W=Y/16$) yields the explicit polynomial parametrization:

$$\begin{aligned}
a(u)&=2(124u^2-3988u+31029),\\
b(u,W)&=-(92u^2-3988u+37365)+W,\\
c(u,W)&=-(92u^2-3988u+37365)-W,\\
d(u)&=-56(u^2-198),\\
e(u)&=4(173u^2-3988u+21327),
\end{aligned} \tag{22}$$

with exact coordinate sum

$$s(u)=a(u)+b(u,W)+c(u,W)+d(u)+e(u)=700u^2-15952u+83724,$$

where (u,W) lies on the smooth genus-1 curve

$$\mathcal{E}_2:\quad W^2=-89200u^4+3979968u^3-68337912u^2+541908504u-1686900843,$$

whose quartic polynomial has non-zero discriminant

$$\Delta(\mathcal{E}_2)=-6135110588211370124157501980003417456 64\neq0,$$

and whose base point lies at the tangent slope $u_0=\frac{55581}{3988}$ with $W_0=\frac{2928874599}{3976036}=\frac{11715498396}{3988^2}$. Under the shift $u=\frac{55581+\tilde{u}}{3988}$ and rescaled ordinate $\tilde{W}=3988^2W$, $\mathcal{E}_2$ becomes the integral quartic $\tilde{W}^2=c_4\tilde{u}^4+c_3\tilde{u}^3+c_2\tilde{u}^2+c_1\tilde{u}+E_0^2$ with exact integer coefficients

$$\begin{aligned}
&c_4=-89200, && c_3=-3959188416, && c_2=-93657652509216,\\
&c_1=-611289683911440576, && E_0=11715498396.
\end{aligned}$$

Proposition 8.2 below maps this model isomorphically onto the integral Weierstrass cubic $E_2:Y_W^2=X_W^3+a_2X_W^2+a_4X_W+a_6$ with

$$\begin{aligned}
a_2&=-374630610036864,\\
a_4&=39506925936543168210437062656,\\
a_6&=-22890842312228182917877567026561967781314 56,
\end{aligned}$$

on which the base point P_0^W already has non-integral X_W-coordinate

$$X_W(P_0^W)=\frac{7436281444943548800}{2401}=\frac{7436281444943548800}{49^2}\notin\mathbb{Z},$$

proving by the Nagell–Lutz Theorem that P_0^W has **infinite order** in $E_2(\mathbb{Q})$ ($\operatorname{rank}E_2(\mathbb{Q})\geq 1$). By Theorem 5.1, the parametrization (22) has degree 1 onto its image curve and avoids zero coordinates outside at most 24 points on $\mathcal{E}_2$. To establish universal 5-adic sector invariance on $\mathcal{E}_2(\mathbb{Q})$, let $(u,W)=(p/q,\tilde{V}/q^2)\in\mathcal{E}_2(\mathbb{Q})$ be any rational point with $\gcd(p,q)=1$:

- If $5\nmid q$ ($u\in\mathbb{Z}_{(5)}$): Modulo 5, the right-hand side of $\mathcal{E}_2$ takes the values $(2,2,1,2,3)$ for $u\equiv(0,1,2,3,4)\pmod 5$. Since 2 and 3 are quadratic non-residues modulo 5, $W^2\equiv 1\pmod 5$ forces $u\equiv 2\pmod 5$, at which $s(u)=700u^2-15952u+83724\equiv 0-2(2)+4\equiv 0\pmod 5$ while $d(u)=-56(u^2-198)\equiv-(4-3)\equiv 4\not\equiv 0\pmod 5$.
- If $5\mid q$ (so $5\nmid p$): Directly from the homogenized coordinate polynomials, $q^2s(p/q)=700p^2-15952pq+83724q^2\equiv 0\pmod 5$ (since $5\mid 700$ and $5\mid q$), whereas $q^2d(p/q)=-56p^2+11088q^2\equiv 4p^2\not\equiv 0\pmod 5$.

In both cases, $v_5(q^2s(p/q))\geq 1>0=v_5(q^2d(p/q))$, so clearing q^2 and dividing by $\gcd(q^2a,\dots,q^2e)$ preserves $5\mid s_{\text{prim}}$ for every rational point on $\mathcal{E}_2(\mathbb{Q})$. Thus $\mathcal{E}_2(\mathbb{Q})$ generates infinitely many primitive solutions with $5\mid s$.

6 General 6-Point Osculating-Parabola Map Around Symmetric Solutions ($b_0=c_0$)

Theorem 6.1 (6-Point Osculating-Parabola Rational Map)**.** *Let $P_0=(a_0,-x_0,-x_0,d_0,e_0)$ be any solution with two equal non-zero coordinates $b_0=c_0=-x_0\neq 0$ (such as Row 1, Row 2, or Row 5 of Table I). For any direction vector $Q=(a_2,0,0,d_2,e_2)$ parameterized by $(a_2:d_2:e_2)\in\mathbb{P}^2(\mathbb{Q})$, consider the 2-plane $\Pi_{(a_2:d_2:e_2)}$ given by*

$$a=a_0x+a_2z,\quad b=-x_0x+y,\quad c=-x_0x-y,\quad d=d_0x+d_2z,\quad e=e_0x+e_2z.$$

Then the plane quartic $C=V\cap\Pi_{(a_2:d_2:e_2)}$ has the equation

$$\left(3x_0^2x^2+y^2\right)^2=9x_0^4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4,\tag{23}$$

where $c_k(a_2,d_2,e_2)$ is homogeneous of degree $4-k$. Since $x_0\neq 0$, define

$$\beta=\frac{c_3}{6x_0^2},\qquad\gamma=\frac{c_2-\beta^2}{6x_0^2},\qquad R_1=c_1-2\beta\gamma,\qquad R_0=c_0-\gamma^2.\tag{24}$$

*Then the parabola $y^2=z(\beta x+\gamma z)$ osculates C at $P_0=(1:0:0)$ with **multiplicity 6**, and whenever $R_1(a_2,d_2,e_2)\neq 0$, the remaining two intersection points have rational x/z and $(y/z)^2$ over $\mathbb{P}^2_{(a_2:d_2:e_2)}$ given by:*

$$\frac{x}{z}=-\frac{R_0(a_2,d_2,e_2)}{R_1(a_2,d_2,e_2)},\qquad\left(\frac{y}{z}\right)^2=\frac{\gamma(a_2,d_2,e_2)R_1(a_2,d_2,e_2)-\beta(a_2,d_2,e_2)R_0(a_2,d_2,e_2)}{R_1(a_2,d_2,e_2)}.\tag{25}$$

In particular, a direction $(a_2:d_2:e_2)\in\mathbb{P}^2(\mathbb{Q})$ yields a rational point on $V(\mathbb{Q})$ whenever the right-hand side of $(y/z)^2$ in (25) is a rational square.

Proof. By definition of β,γ,R_1,R_0 (valid since $x_0\neq 0$), we have the algebraic identity

$$9x_0^4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4=\left(3x_0^2x^2+\beta xz+\gamma z^2\right)^2+z^3(R_1x+R_0z).$$

Equating $(3x_0^2x^2+y^2)^2$ to the right-hand side and imposing $3x_0^2x^2+y^2=3x_0^2x^2+\beta xz+\gamma z^2$ (which cancels $3x_0^2x^2$ to give $y^2=\beta xz+\gamma z^2$) forces $z^3(R_1x+R_0z)=0$. The six roots $z^3=0$ (each giving $y^2=0$, multiplicity 2) account for 6 intersection points at $P_0=(1:0:0)$, while for $R_1\neq 0$ the linear factor $R_1x+R_0z=0$ yields $\frac{x}{z}=-\frac{R_0}{R_1}$ and $\left(\frac{y}{z}\right)^2=\beta\frac{x}{z}+\gamma=\frac{\gamma R_1-\beta R_0}{R_1}$. □

7 Rational Elimination of d and the Separation-of-Variables Surface

Theorem 7.1 (Rational Elimination of d in Terms of Free Parameters (a,x,e))**.** *In the Three-Squares Identity (8), setting the second and third squares equal via $d^2+de+e^2=4(x^2+xz+z^2)-2a(x+z)+a^2$ (with $z=-\frac{d+e}{2}$) cancels the d^2 term identically and* ***solves linearly and rationally for*** *d in terms of (a,x,e) on the affine chart $a+e-2x\neq 0$:*

$$d=-\frac{a^2+ae-2ax-2ex+4x^2}{a+e-2x},\qquad b=-x+y,\qquad c=-x-y. \tag{26}$$

Under (26), setting $a=-m^2x$ reduces the 5-variable fourth-power equation (for $e\neq(m^2+2)x$) to a binary quartic in (e,x) for the norm y^2+3x^2, whose leading e^4 coefficient is $(2mx)^2=4m^2x^2$ (a perfect square for every $m\in\mathbb{Q}$):

$$\Big[(y^2+3x^2)\Big(e-(m^2+2)x\Big)\Big]^2=\left(2mxe^2-m(m^2+2)x^2e-\frac{m(m^2+5)(m^4+4)}{4}x^3\right)^2 + R_1(m)x^5e+R_0(m)x^6, \tag{27}$$

where

$$R_1(m)=\frac{3}{2}m^2(m^2+1)(m^2+2)(m^4+4), \tag{28}$$

$$R_0(m)=-\frac{1}{16}m^2(m^2+1)(m^4+4)\left(m^6+25m^4+68m^2+100\right), \tag{29}$$

yielding for every $m\in\mathbb{Q}\setminus\{0\}$ (where $R_1(m)>0$) the rational specialization $\frac{e}{x}=-\frac{R_0(m)}{R_1(m)}=\frac{m^6+25m^4+68m^2+100}{24(m^2+2)}$ for the norm y^2+3x^2.

Proof. Substituting $z=-\frac{d+e}{2}$ into $4(x^2+xz+z^2)-2a(x+z)+a^2-(d^2+de+e^2)=0$ cancels d^2+de+e^2 against $4z^2-de=(d+e)^2-de$ and leaves $(a-2x+e)d+(a^2+ae-2ax-2ex+4x^2)=0$, giving (26) when $a+e-2x\neq 0$. Substituting

$$z=-\frac{d+e}{2}=\frac{a^2-2ax+4x^2-e^2}{2(a+e-2x)}$$

into $-a(a^3-8axz+16xz(x+z))(a+e-2x)^2$ with $a=-m^2x$ yields a degree-4 polynomial in e with leading term $-a(4xe^4)=4m^2x^2e^4=(2mxe^2)^2$. Completing the square in e gives the stated $R_1(m)$ and $R_0(m)$, where the common factor $m^2(m^2+1)(m^4+4)$ arises because the quadratic in z has zero discriminant when $a=x$ ($m^2=-1$) or $a^2+4x^2=0$ ($m^4=-4$). □

Remark 7.2 (Genus-10 Hyperelliptic Obstruction on the Linear Specialization $e/x=-R_0(m)/R_1(m)$)**.** Substituting $\frac{e}{x}=\frac{m^6+25m^4+68m^2+100}{24(m^2+2)}$ into (27) gives $\frac{e}{x}-(m^2+2)=\frac{m^6+m^4-28m^2+4}{24(m^2+2)}$ and expresses $\left(\frac{y}{x}\right)^2=\frac{y^2+3x^2}{x^2}-3$ as

$$12(m^2+2)(m^6+m^4-28m^2+4)\left(\frac{y}{x}\right)^2$$
$$=\pm m\left(m^{12}-34m^{10}-235m^8-480m^6-72m^4-1376m^2-560\right)-36(m^2+2)(m^6+m^4-28m^2+4).$$

Clearing the squarefree degree-8 denominator shows that $(y/x)\in\mathbb{Q}$ requires a rational point on $Y^2=P_{21}(m)$, where $P_{21}(m)$ is a squarefree polynomial of degree 21 in m. This defines a smooth hyperelliptic curve of genus $g=\lfloor(21-1)/2\rfloor=10$, which by Faltings' Theorem has only finitely many rational points (and none of small height).

Theorem 7.3 (General Separation-of-Variables Surface)**.** *For any solution (a,b,c,d,e), set $b=-x+y$, $c=-x-y$, and $d=2x+p$ (so $p=b+c+d$ and $s=a+p+e$). Then the $16x^4$ terms cancel between b^4+c^4 and d^4, and the variables a and x separate completely:*

$$2\left(y^2+3x^2\right)^2=\left[(a+p+e)^4-a^4-p^4-e^4\right]-\left[(2x+p)^4-(2x)^4-p^4\right]. \tag{30}$$

In particular, on the hyperplane $p=2e$ (i.e., $s-a=3e$, which is satisfied by ***Rows 3, 4, 16, and 17 of Table I****: directly with a as the odd coordinate in Rows 4, 16, and 17, and with $a=-9, e=8$ in Row 3), (30) simplifies to the separated cubic surface:*

$$\left(y^2+3x^2\right)^2=e\left(6a^3+27ea^2+54e^2a+32e^3-32x^3-48ex^2-32e^2x\right). \tag{31}$$

Proof. Since $b^4+c^4+d^4=2(y^2+3x^2)^2-16x^4+(2x+p)^4=2(y^2+3x^2)^2+\left[(2x+p)^4-(2x)^4-p^4\right]+p^4$, subtracting from $s^4-a^4-e^4=(a+p+e)^4-a^4-e^4$ immediately yields (30). Setting $p=2e$ and dividing by 2 gives (31). □

8 Universal Tangent-Slope Shift, Integral Weierstrass Models, and Closed-Form Rational Recurrence

Theorem 8.1 (Universal Tangent-Slope Shift and Closed-Form Recurrence Step)**.** *In Theorem 5.1, when the conic $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ is parameterized by $U-U_0=t(x-x_0)$, the base solution (x_0,y_0,U_0) with $U(t_0)=+U_0>0$ is located at the tangent slope*

$$t_0=\frac{2\mathcal{A}x_0+\mathcal{B}}{2U_0}. \tag{32}$$

Shifting the parameter via $t=t_0+\kappa u$ (for any scaling constant $\kappa\in\mathbb{Q}^\times$, and rescaling Y accordingly) places the base solution at $u=0$ on a quartic curve of the form

$$\mathcal{E}:\quad Y^2=c_4u^4+c_3u^3+c_2u^2+c_1u+E_0^2\qquad(E_0>0). \tag{33}$$

The osculating parabola $Y(u)=E_0+m_1u+m_2u^2$ at $(0,E_0)$ with

$$m_1=\frac{c_1}{2E_0},\qquad m_2=\frac{4E_0^2c_2-c_1^2}{8E_0^3}$$

cancels the u^0,u^1,u^2 coefficients of $Y(u)^2-\mathcal{E}(u)=0$ identically and, whenever $m_2^2-c_4\neq0$ (i.e., $-64E_0^6c_4+16E_0^4c_2^2-8E_0^2c_1^2c_2+c_1^4\neq0$) and $2m_1m_2-c_3\neq0$ (i.e., $8E_0^4c_3-4E_0^2c_1c_2+c_1^3\neq0$), yields the distinct non-zero closed-form next rational point $(u_{\text{new}},Y_{\text{new}})$:

$$u_{\text{new}}=\frac{8E_0^2\left(8E_0^4c_3-4E_0^2c_1c_2+c_1^3\right)}{-64E_0^6c_4+16E_0^4c_2^2-8E_0^2c_1^2c_2+c_1^4},\qquad Y_{\text{new}}=E_0+m_1u_{\text{new}}+m_2u_{\text{new}}^2. \tag{34}$$

Proof. Differentiating $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ at (x_0,U_0) gives $2U_0\frac{dU}{dx}=2\mathcal{A}x_0+\mathcal{B}$, so the tangent line at (x_0,U_0) has slope $t_0=\frac{2\mathcal{A}x_0+\mathcal{B}}{2U_0}$, where the second intersection point on the conic coincides with $(x_0,+U_0)$ and $y^2(t_0)=U_0-3x_0^2=+y_0^2>0$. Expanding $(E_0+m_1u+m_2u^2)^2-(c_4u^4+c_3u^3+c_2u^2+c_1u+E_0^2)$ with the given m_1,m_2 leaves $u^3[(m_2^2-c_4)u+(2m_1m_2-c_3)]=0$, whose root $u_{\text{new}}=-\frac{2m_1m_2-c_3}{m_2^2-c_4}$ simplifies directly to (34) and is non-zero whenever both $m_2^2-c_4\neq0$ and $2m_1m_2-c_3\neq0$. □

To prove that the rational point $(0,E_0)$ on (33) has infinite order, we establish the explicit birational isomorphism to an integral Weierstrass model and apply the Nagell–Lutz Theorem.

Proposition 8.2 (Integral Weierstrass Isomorphism and Nagell–Lutz Non-Torsion Criterion)**.** *Given any quartic curve $W^2 = g(t) \in \mathbb{Z}[t]$ with rational base point $t_0 = p_0/q_0$ ($\gcd(p_0, q_0) = 1$) and $W(t_0) = W_0 \in \mathbb{Q}_{>0}$, substituting $t = \frac{p_0+u}{q_0}$ and $Y = q_0^2 W$ clears q_0^4 to produce a quartic of the form (33) in which all coefficients $c_1, c_2, c_3, c_4 \in \mathbb{Z}$ and $E_0 = q_0^2 W_0 \in \mathbb{Z}_{>0}$ are integers. Suppose that $c_1, c_2, c_3, c_4, E_0 \in \mathbb{Z}$ and that the quartic polynomial in (33) has non-zero discriminant. Choosing $Q_0 = (0, -E_0) \in \mathcal{E}(\mathbb{Q})$ as the identity element $\mathcal{O}$, the birational transformation*

$$X_W = \frac{8E_0(E_0 + m_1 u - Y)}{u^2}, \qquad Y_W = 8E_0\left[2\left(\frac{X_W^2}{64E_0^2} - c_4\right)u - \frac{m_1}{4E_0}X_W - c_3\right] \tag{35}$$

with inverse map

$$u = \frac{4E_0\left(Y_W + \frac{c_1}{E_0}X_W + 8E_0 c_3\right)}{X_W^2 - 64E_0^2 c_4} = \frac{4\left(E_0 Y_W + c_1 X_W + 8E_0^2 c_3\right)}{X_W^2 - 64E_0^2 c_4}, \qquad Y = E_0 + m_1 u - \frac{X_W}{8E_0}u^2$$

maps $\mathcal{E}$ isomorphically onto the ***integral Weierstrass cubic***

$$E: \quad Y_W^2 = X_W^3 + a_2 X_W^2 + a_4 X_W + a_6, \tag{36}$$

with integer coefficients

$$a_2 = 4c_2 \in \mathbb{Z}, \qquad a_4 = 16(c_1 c_3 - 4E_0^2 c_4) \in \mathbb{Z}, \qquad a_6 = 64\left(E_0^2 c_3^2 - 4E_0^2 c_2 c_4 + c_1^2 c_4\right) \in \mathbb{Z}.$$

Under (35), the base point $P_0 = (0, +E_0) \in \mathcal{E}(\mathbb{Q})$ maps to

$$P_0^W = \left(\frac{c_1^2 - 4E_0^2 c_2}{E_0^2},\ \frac{c_1(4E_0^2 c_2 - c_1^2) - 8E_0^4 c_3}{E_0^3}\right) \in E(\mathbb{Q}), \tag{37}$$

and the hyperelliptic conjugate $(u_{\text{new}}, -Y_{\text{new}})$ of the osculating-parabola point (34) maps to the doubled point $2P_0^W \in E(\mathbb{Q})$. By the Nagell–Lutz Theorem [5, Ch. VIII, Cor. 7.2], if either $E_0 \nmid c_1$ (so $X_W(P_0^W) \notin \mathbb{Z}$) or $X_W(2P_0^W) \notin \mathbb{Z}$, then P_0^W is non-torsion and has ***infinite order*** *in $E(\mathbb{Q})$ ($\operatorname{rank} E(\mathbb{Q}) \geq 1$).*

Proof. Substituting $Y = E_0 + m_1 u - \frac{X_W}{8E_0}u^2$ into $Y^2 - (c_4 u^4 + c_3 u^3 + c_2 u^2 + c_1 u + E_0^2) = 0$ and dividing by u^2 gives the quadratic equation in u

$$\left(\frac{X_W^2}{64E_0^2} - c_4\right)u^2 - \left(\frac{m_1}{4E_0}X_W + c_3\right)u + \left(m_1^2 - c_2 - \frac{X_W}{4}\right) = 0,$$

whose discriminant in u, multiplied by $64E_0^2$, simplifies via $2E_0 m_1 = c_1$ directly to $Y_W^2 = X_W^3 + a_2 X_W^2 + a_4 X_W + a_6$ with $Y_W = 8E_0\left[2\left(\frac{X_W^2}{64E_0^2} - c_4\right)u - \frac{m_1}{4E_0}X_W - c_3\right]$, while solving this linear relation in (35) for u using $\frac{m_1}{4E_0} = \frac{c_1}{8E_0^2}$ yields the stated inverse formula for u. Because $c_1, c_2, c_3, c_4, E_0 \in \mathbb{Z}$, all three coefficients a_2, a_4, a_6 are integers. Along $Y = E_0 + m_1 u + m_2 u^2 + O(u^3)$ as $u \to 0$, $X_W \to -8E_0 m_2 = \frac{c_1^2 - 4E_0^2 c_2}{E_0^2}$ and $Y_W \to 8E_0(2m_1 m_2 - c_3)$, giving (37). Direct symbolic evaluation confirms $X_W(u_{\text{new}}, -Y_{\text{new}}) = \frac{8E_0(E_0 + m_1 u_{\text{new}} + Y_{\text{new}})}{u_{\text{new}}^2} = X_W(2P_0^W)$. Since $a_2, a_4, a_6 \in \mathbb{Z}$, the Nagell–Lutz Theorem guarantees that every non-zero torsion point in $E(\mathbb{Q})$ has coordinates in $\mathbb{Z}^2$; thus non-integrality of $X_W(P_0^W)$ or $X_W(2P_0^W)$ proves infinite order. □

9 Explicit Parametrization for the $5 \nmid (a+b+c+d+e)$ Sector ($e_0 = \frac{3}{5}$)

We now prove rigorously that the second table ($5 \nmid (a+b+c+d+e)$) also contains infinitely many primitive solutions by constructing an explicit polynomial parametrization from Row 4 of Table II and establishing infinite order ($\operatorname{rank} E_{3/5}(\mathbb{Q}) \geq 1$), generic injectivity, zero-coordinate avoidance, and universal 5-adic sector invariance.

Theorem 9.1 (Explicit Polynomial Parametrization and Infinitude for the $5 \nmid s$ Sector)**.** *Define (a,b,c,d,e) in terms of (t,W) by*

$$\begin{aligned} a(t) &= 180t^2 - 51730t + 5927700, \\ b(t,W) &= \frac{-5(9t^2 - 10346t + 1434858) + 5W}{2}, \\ c(t,W) &= \frac{-5(9t^2 - 10346t + 1434858) - 5W}{2}, \\ d(t) &= 490(t^2 - 9234), \\ e(t) &= -74t^2 - 31038t + 5237208, \end{aligned} \tag{38}$$

with exact coordinate sum

$$s(t) = a(t) + b(t,W) + c(t,W) + d(t) + e(t) = 551t^2 - 31038t - 534042. \tag{39}$$

Then:

1. ***Algebraic Identity:*** *The fourth-power relation $a^4+b^4+c^4+d^4+e^4=s^4$ holds identically whenever (t,W) lies on the smooth genus-1 quartic curve*

$$\mathcal{E}_{3/5}: \quad W^2 = -20935t^4 + 6630540t^3 - 398601480t^2 + 33002726904t - 4412112725340, \tag{40}$$

whose quartic polynomial has non-zero discriminant

$$\Delta(\mathcal{E}_{3/5}) = -20588652091026189210748698255755614541702263267000320 0 \neq 0.$$

2. ***Infinite Order ($\operatorname{rank} E_{3/5}(\mathbb{Q}) \geq 1$) and Explicit 44-Digit Solution:*** *At $t_0 = \frac{108426}{739}$, (38) recovers Row 4 of Table II, $(180, 335, -380, 490, -74)$ with $s = 551$ and $W_0 = \frac{960004211166}{739^2}$. Under the integral shift $t = \frac{108426+\tilde{u}}{739}$ and $\tilde{W} = 739^2 W$ of Proposition 8.2 (which yields integer quartic coefficients $c_4 = -20935$, $c_3 = -4179624180$, $c_2 = -100528467920760$, $c_1 = 32187524610126004056$, and $E_0 = 960004211166$), $\mathcal{E}_{3/5}$ maps isomorphically onto the integral Weierstrass cubic $E_{3/5}: Y_W^2 = X_W^3 + a_2 X_W^2 + a_4 X_W + a_6$ with*

$$\begin{aligned} a_2 &= -402113871683040, \\ a_4 &= -917700721259268291448218474240, \\ a_6 &= -85426882193031845449484208829499884100173824 0, \end{aligned}$$

on which the base point P_0^W has non-integral coordinate

$$X_W(P_0^W) = \frac{31210814234217589104}{20449} = \frac{31210814234217589104}{143^2} \notin \mathbb{Z},$$

*so P_0^W **has infinite order in** $\mathcal{E}_{3/5}(\mathbb{Q})$ by the Nagell–Lutz Theorem. In terms of $t = \frac{108426}{739} + u$, the recurrence step (34) yields*

$$u_{\text{new}} = \frac{\tilde{u}_{\text{new}}}{739} = \frac{342968223159767136957819393 6}{616691523480958080194046 23},$$

which upon clearing denominators and the greatest common divisor produces the 44-digit primitive solution in $(\mathbb{Z} \setminus \{0\})^5$ with $s \equiv 2 \not\equiv 0 \pmod 5$:

$$\begin{aligned} a &= 29355898637232699168520596016080362847497 00, \\ b &= 69884880500131990163419576120217586951366 55, \\ c &= -54841817257585917918331941622823931981005 40, \\ d &= 16115178756364146661235579964149828689444 810, \\ e &= -42242839055584353812048367971479717423725 38, \\ s &= 16330791038783588421391566218349258728858087. \end{aligned}$$

3. ***Generic Injectivity, Zero-Coordinate Avoidance, and Universal 5-Adic Sector Invariance:*** *The rational map $\Phi : \mathcal{E}_{3/5} \dashrightarrow V \subset \mathbb{P}^4$ has degree 1 onto its image curve and avoids zero coordinates outside at most 24 points on $\mathcal{E}_{3/5}$. Every rational point $(t, W) \in \mathcal{E}_{3/5}(\mathbb{Q})$ has 5-integral coordinates $t, W \in \mathbb{Z}_{(5)}$, and every resulting primitive integer solution satisfies $5 \nmid s_{\text{prim}}$ (with $625 \mid (s_{\text{prim}} - e_{\text{prim}})$). Thus $\mathcal{E}_{3/5}(\mathbb{Q})$ generates infinitely many primitive solutions with $5 \nmid (a+b+c+d+e)$.*

Proof. In Row 4 of Table II, $(a_0, b_0, c_0, d_0, e_0^*) = (180, 335, -380, 490, -74)$ with $s = 551$ satisfies

$$\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0} = \frac{180+335-380}{180+335-380+490} = \frac{135}{625} = \frac{27}{125} = \left(\frac{3}{5}\right)^3 \implies e_0 = \frac{3}{5}.$$

Setting $z = 625$, $x = \frac{5}{2}X$, and $y = \frac{5}{2}Y$ on $H_{3/5}$ gives $a = 5X + 135$, $b = \frac{-5X+5Y}{2}$, $c = \frac{-5X-5Y}{2}$, $d = 490$, $e = 3X - 101$, and reduces (13) to

$$(3X^2 + Y^2)^2 = 147744X^2 + 21628032X + 221539312,$$

with base point $(X_0, Y_0) = (9, 143)$ and $U_0 = 3(9)^2 + 143^2 = 20692$. Parameterizing the conic via $3X^2 + Y^2 = 20692 + 4t(X - 9)$ gives $X(t) = \frac{9t^2-10346t+1434858}{t^2-9234}$, and clearing $t^2 - 9234$ yields (38), the sum (39), and the nonsingular quartic $\mathcal{E}_{3/5}$ in (40). By Theorem 5.1, Φ has degree 1 onto its image curve (since $X = -125\frac{b+c}{a+b+c+d}$ and $Y = 125\frac{b-c}{a+b+c+d}$ recover $t = \frac{3X^2+Y^2-20692}{4(X-9)}$ and $W = \frac{Y(t^2-9234)}{5}$ uniquely), and each of the six non-zero functions a, b, c, d, e, s has degree 4 on $\mathcal{E}_{3/5}$, so $abcdes = 0$ defines at most 24 points on $\mathcal{E}_{3/5}$.

To prove universal 5-adic sector invariance in part (3), let $(t, W) \in \mathcal{E}_{3/5}(\mathbb{Q})$ be any rational point. Suppose first that $v_5(t) = -k < 0$ with $k \geq 1$. Since $20935 = 5 \cdot 4187$ with $5 \nmid 4187$, we have $v_5(-20935t^4) = 1 - 4k$, whereas the remaining four terms on the right-hand side of (40) have 5-adic valuations $\geq 1-3k, 1-2k, -k, 1$, all strictly greater than $1 - 4k$. By the strict triangle inequality for v_5, $v_5(W^2) = 1 - 4k$, which is odd, contradicting $v_5(W^2) = 2v_5(W) \in 2\mathbb{Z}$. Hence **every rational point on $\mathcal{E}_{3/5}(\mathbb{Q})$ is 5-integral**: $v_5(t) \geq 0$ and $v_5(W) \geq 0$. Because $t, W \in \mathbb{Z}_{(5)}$, the polynomials $a(t), b(t, W), c(t, W), d(t)$ in (38) all have 5-adic valuation ≥ 1 (with $s(t) - e(t) = a + b + c + d = 625(t^2 - 9234)$), whereas

$$s(t) \equiv e(t) \equiv t^2 + 2t - 2 \equiv (t+1)^2 + 2 \pmod 5.$$

Since the quadratic residues in $\mathbb{F}_5$ are $0, 1, 4$, we have $(t+1)^2 + 2 \in \{2, 3, 1\} \pmod 5$, which is never $0 \pmod 5$. Thus $v_5(e(t)) = v_5(s(t)) = 0 < 1 \leq \min(v_5(a), v_5(b), v_5(c), v_5(d))$, so clearing the 5-adic unit denominator and dividing out the greatest common divisor preserves $5 \nmid s_{\text{prim}}$ for every rational point on $\mathcal{E}_{3/5}(\mathbb{Q})$. Since $\Phi : \mathcal{E}_{3/5} \dashrightarrow V$ has degree 1 onto its image curve and P_0^W has infinite order, $\mathcal{E}_{3/5}(\mathbb{Q})$ generates infinitely many primitive solutions with $5 \nmid s$. □

10 General Non-Even Osculating Conic Around an Arbitrary Solution ($b_0 \neq c_0$)

We can drop the symmetry requirement $b_0 = c_0$ of Theorem 6.1 and construct a 5-point osculating conic around any rational solution $P_0 \in V(\mathbb{Q})$ with $b_0 \neq c_0$ and $b_0 + c_0 \neq 0$.

Theorem 10.1 (Non-Even Osculating Conic System)**.** *Let $P_0 = (a_0, b_0, c_0, d_0, e_0) \in V(\mathbb{Q})$ be any solution with $b_0 \neq c_0$ and $b_0 + c_0 \neq 0$, normalized by scaling so that $b_0 + c_0 = -1$, and set $y_0 = \frac{b_0-c_0}{2} \neq 0$. On the 2-plane $\Pi_{(a_2:d_2:e_2)}$ given by*

$$a = a_0x + a_2z, \quad b = -\frac{1}{2}x + y, \quad c = -\frac{1}{2}x - y, \quad d = d_0x + d_2z, \quad e = e_0x + e_2z,$$

the plane quartic $C = V \cap \Pi_{(a_2:d_2:e_2)}$ is

$$\left(\frac{3}{4}x^2+y^2\right)^2 = c_4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4, \qquad c_4=\left(\frac{3}{4}+y_0^2\right)^2,$$

containing both $P_0=(1:y_0:0)$ and $P_2=(1:-y_0:0)$ on $z=0$. Consider the conic in $\Pi_{(a_2:d_2:e_2)}$ passing through P_0 (at $t=\infty$) and P_2 (at $t=0$):

$$x(t)=t^2+x_1t+x_0, \qquad y(t)=y_0t^2+y_1t-y_0x_0, \qquad z(t)=t. \tag{41}$$

*The conditions for the coefficients of t^7, t^6, t^5 on C to vanish (so the conic osculates C to order 4 at P_0 while also passing through P_2) **decouple into a triangular linear system in** (y_1, x_0, x_1) (valid whenever $2c_2y_0^2(4y_0^2+3)^2-3c_3^2(4y_0^2+1)\neq 0$):*

$$y_1 = y_0x_1+\frac{c_3}{y_0(4y_0^2+3)}, \tag{42}$$

$$x_0 = \frac{3c_3^2(4y_0^2+1)-2c_2y_0^2(4y_0^2+3)^2}{4y_0^4(4y_0^2+3)^3}, \tag{43}$$

$$x_1 = \frac{2c_1y_0^4(4y_0^2+3)^4-6c_2c_3y_0^2(4y_0^2+1)(4y_0^2+3)^2+c_3^3(112y_0^4+48y_0^2+9)}{y_0^2(4y_0^2+3)^2\left[2c_2y_0^2(4y_0^2+3)^2-3c_3^2(4y_0^2+1)\right]}. \tag{44}$$

With these explicit rational formulas, 5 of the 8 intersection points of the conic with C lie at P_0 and P_2, leaving a residual cubic $d_4t^3+d_3t^2+d_2t+d_1=0$ with $d_1=-2y_0(4y_0^2+3)x_0^3y_1$.

Proof. Substituting (41) into $(\frac{3}{4}x^2+y^2)^2-(c_4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4)$ yields a polynomial in t whose t^8 and t^0 coefficients vanish identically. Its t^7 coefficient is $-c_3-y_0^2(4y_0^2+3)x_1+y_0(4y_0^2+3)y_1=0$, giving (42). Substituting (42) into the t^6 coefficient cancels all x_1^2 and x_1 terms identically, leaving a linear equation in x_0 whose unique solution is (43). Finally, substituting both (42) and (43) into the t^5 coefficient cancels all x_1^3 and x_1^2 terms identically, leaving a linear equation in x_1 whose unique solution is (44). The coefficient of t^1 is $-x_0^3(c_3+y_0^2(4y_0^2+3)x_1+y_0(4y_0^2+3)y_1)$, which upon substituting $c_3+y_0^2(4y_0^2+3)x_1=y_0(4y_0^2+3)y_1$ from (42) equals $-2y_0(4y_0^2+3)x_0^3y_1$. □

11 Depressed-Cubic Reduction on the Separated Surface for Arbitrary $(e,p)\in\mathbb{Q}^2$

Theorem 11.1 (2-Parameter Depressed-Cubic Surface Family)**.** *In the general separated equation (30) of Theorem 7.3,*

$$2\left(y^2+3x^2\right)^2 = \left[(a+p+e)^4-a^4-p^4-e^4\right]-\left[(2x+p)^4-(2x)^4-p^4\right],$$

setting $a=2x+q$ with

$$q=-\frac{e(e+2p)}{2(e+p)} \tag{45}$$

eliminates both the x^4 and x^2 terms identically for every pair $(e,p)\in\mathbb{Q}^2$ with $e+p\neq 0$, reducing the equation to the 2-parameter family of depressed-cubic surfaces:

$$\left(y^2+3x^2\right)^2 = 16ex^3+\frac{e^2(e^2+4ep+6p^2)}{e+p}x+\frac{e^2(4p^4+4ep^3-e^2p^2-4e^3p-2e^4)}{4(e+p)^2}, \tag{46}$$

where (a,b,c,d,e) is reconstructed rationally via $a=2x-\frac{e(e+2p)}{2(e+p)}$, $b=-x+y$, $c=-x-y$, $d=2x+p$, and $e=e$.

Proof. Substituting $a=2x+q$ into the right-hand side of (30) and dividing by 2 yields a cubic polynomial in x with leading term $16ex^3$ and x^2 coefficient $12(e^2+2ep+2eq+2pq)=12[e(e+2p)+2(e+p)q]$. Setting $q=-\frac{e(e+2p)}{2(e+p)}$ (for $e+p\neq 0$) makes the x^2 coefficient vanish identically and simplifies the x^1 and x^0 coefficients to the stated rational expressions. □

12 The Centered Double-Depressed-Cubic Identity (Global Canonical Form)

We now unify all previous reductions into a global canonical form for the threefold V.

Theorem 12.1 (Centered Double-Depressed-Cubic Canonical Form)**.** *For any (a,b,c,d,e) with sum $s=a+b+c+d+e$, define the centered coordinates*

$$u=a+b+c,\quad v=a-\frac{u}{2}=\frac{a-b-c}{2},\quad r=\frac{s+d}{2}=d+\frac{u+e}{2},\quad x=\frac{2v-u}{4}=-\frac{b+c}{2},\quad y=\frac{b-c}{2}, \tag{47}$$

with inverse linear reconstruction

$$a=v+\frac{u}{2},\quad b=-x+y,\quad c=-x-y,\quad d=r-\frac{u+e}{2},\quad e=e,\quad s=r+\frac{u+e}{2}.$$

*Then the Diophantine equation $a^4+b^4+c^4+d^4+e^4=s^4$ is **identically equivalent** to the Centered Double-Depressed-Cubic Identity:*

$$2\left(y^2+3x^2\right)^2=(u+e)r\left(4r^2+(u+e)^2\right)-uv\left(4v^2+u^2\right)-e^4. \tag{48}$$

Consequently:

1. *Because (48) is already depressed in both r and v (containing no r^2 or v^2 terms), setting $r=mv$ preserves the absence of the v^2 term and gives a 3-parameter family of depressed cubics in v for all $(u,e,m)\in\mathbb{Q}^3$:*

$$2\left(y^2+3x^2\right)^2=4\left[(u+e)m^3-u\right]v^3+\left[(u+e)^3m-u^3\right]v-e^4.$$

2. *Additionally imposing $\frac{u}{u+e}=m^3$ (i.e., $u=\frac{m^3}{1-m^3}e$ for $m\in\mathbb{Q}\setminus\{1\}$) eliminates the v^3 term as well (so both the v^3 and v^2 coefficients are zero), leaving a purely linear right-hand side in v:*

$$2\left(y^2+3x^2\right)^2=\frac{m(1-m^8)}{(1-m^3)^3}e^3v-e^4.$$

3. ***All-Integer Canonical Coordinates for Primitive Solutions:** By the 2-Adic Parity Law (Proposition 0.1(2)), every primitive solution in $(\mathbb{Z}\setminus\{0\})^5$ has a unique odd coordinate and four even coordinates. Placing the unique odd coordinate in d and the four even coordinates in (a,b,c,e) ensures that $u\in2\mathbb{Z}$, $e\in2\mathbb{Z}$, and $(u,v,r,x,y,e)\in\mathbb{Z}^6$ **are all exact integers** for every primitive solution.*

Proof. Since $s=r+\frac{u+e}{2}$ and $d=r-\frac{u+e}{2}$, the even powers r^4 and r^2 cancel in s^4-d^4, leaving

$$s^4-d^4=\left(r+\frac{u+e}{2}\right)^4-\left(r-\frac{u+e}{2}\right)^4=(u+e)r\left(4r^2+(u+e)^2\right).$$

Likewise, since $a=v+\frac{u}{2}$ and $-(b+c)=2x=v-\frac{u}{2}$, we have $a^4-(b+c)^4=uv(4v^2+u^2)$, while $b^4+c^4=2(y^2+3x^2)^2-(b+c)^4$. Adding $a^4+b^4+c^4=2(y^2+3x^2)^2+uv(4v^2+u^2)$ to d^4+e^4 and equating to s^4 yields (48). When d is odd and a,b,c,e are even, s is odd, so $u=a+b+c\in2\mathbb{Z}$, $v=a-u/2\in\mathbb{Z}$, $r=(s+d)/2\in\mathbb{Z}$, $x=-(b+c)/2\in\mathbb{Z}$, and $y=(b-c)/2\in\mathbb{Z}$. □

13 The Paired-Solution Surface ($u=2e$) and Weierstrass Fibrations

Theorem 12.1 explains why several solutions in Tables I and II occur in **pairs** sharing identical values of e and $s-d=u+e$:

- **Pair 1 ($u=16, e=8, s-d=24$):** Row 3 ($s=15, d=-9$) has $(r,v,x,y)=(3,-2,-5,9)$, and Row 4 ($s=135, d=111$) has $(r,v,x,y)=(123,18,5,-97)$.
- **Pair 2 ($u=208, e=104, s-d=312$):** Row 16 ($s=435, d=123$) has $(r,v,x,y)=(279,90,-7,361)$, and Row 17 ($s=515, d=203$) has $(r,v,x,y)=(359,40,-32,426)$.
- **Pair 3 in Table II ($u=695, e=-70, s-d=625$):** Placing the non-5-multiple coordinate in d, Row 2 of Table II ($s=413, d=-212$) has $(r,v,x,y)=(\frac{201}{2},-\frac{195}{2},-\frac{445}{2},-\frac{335}{2})$, and Row 3 of Table II ($s=507, d=-118$) has $(r,v,x,y)=(\frac{389}{2},-\frac{995}{2},-\frac{845}{2},-\frac{55}{2})$.

Proposition 13.1 (The Parameter-Free Paired-Solution Surface $u=2e$ and Its Elliptic Quotient)**.** *Both Pair 1 and Pair 2 satisfy $u=2e$ ($s-d=3e$) with $e\neq 0$. Substituting $u=2e$ into (48) and normalizing by e via $(R,V,Y)=(\frac{r}{e},\frac{v}{e},\frac{y}{e})$ collapses all four solutions onto the single parameter-free depressed-cubic surface*

$$2\left(Y^2+\frac{3}{4}(V-1)^2\right)^2=12R^3+27R-8V^3-8V-1, \tag{49}$$

*at $(R,V,Y)=(\frac{3}{8},-\frac{1}{4},\frac{9}{8})$ (Row 3), $(\frac{123}{8},\frac{9}{4},\frac{97}{8})$ (Row 4), $(\frac{279}{104},\frac{45}{52},\frac{361}{104})$ (Row 16), and $(\frac{359}{104},\frac{5}{13},\frac{213}{52})$ (Row 17). Because $X=\frac{V-1}{2}$ is independent of R, fixing any $V=V_0\in\mathbb{Q}$ defines a smooth plane quartic fiber in (R,Y) of **genus 3** (for generic $V_0\in\mathbb{Q}$), together with a 2-to-1 quotient map $(R,Y)\mapsto(R,W)$ via $W=Y^2+\frac{3}{4}(V_0-1)^2$ onto the Weierstrass cubic*

$$2W^2=12R^3+27R-(8V_0^3+8V_0+1)\qquad \text{in } (R,W),$$

*which is **smooth of genus 1 for every** $V_0\in\mathbb{Q}$ because the cubic $12R^3+27R-C(V_0)$ has strictly positive derivative $36R^2+27>0$ on $\mathbb{R}$ and strictly negative discriminant $-4(12)(27)^3-27(12)^2C(V_0)^2<0$. A rational point (R,W) on this elliptic quotient cubic lifts to a rational point (R,V_0,Y) on the genus-3 fiber (49) if and only if $W-\frac{3}{4}(V_0-1)^2$ is a rational square.*

14 Explicit Pencils on the Four-Squares Surface S_0 ($2a+b+c=0$)

The Four-Squares Surface $S_0\subset\mathbb{P}^3$ of Theorem 2.2,

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2+x^2(x+4z)^2=\left(3x^2+2xz+4z^2\right)^2,$$

contains Row 1 at $(x,y,z,w)=(-3,\pm1,-1,\pm3)$, Row 21 (with $(a,b,c,d,e)=(56,-268,156,619,62)$, giving $(x,y,z,w)=(56,\pm212,-\frac{681}{2},\pm\frac{557}{2})$ and projectively represented after scaling by 2 at $(x:y:z:w)=(112:\pm424:-681:\pm557)$, which corresponds to the doubled tuple of sum 1250), and the boundary points $(0:0:1:\pm1)$.

Proposition 14.1 (Two Explicit Pencils of Curves on S_0)**.**

1. ***The Pencil $w=z+tx$ Through $(0:0:1:1)$:*** *Substituting $w=z+tx$ (so $d=tx$ and $e=-2z-tx$) into S_0 cancels the z^4 term identically and factors out x, leaving (for $t\neq1$) the cubic bundle in z:*

$$\left(y^2+3x^2\right)^2=x\left[(8-t^4)x^3-4(t^3-1)x^2z-12(t^2-1)xz^2-16(t-1)z^3\right], \tag{50}$$

which passes through Row 1 at $t=-\frac{4}{3}$ and $t=\frac{2}{3}$ with $(x,y,z)=(-3,\pm1,-1)$.

2. ***The Even Biquadratic Pencil*** $w=ky$***:*** *Because S_0 involves only y^2 and w^2, setting $w=ky$ cuts out a plane quartic curve $C_k\subset S_0$ in $(x:y:z)$ that is a* ***smooth plane quartic of genus 3*** *for generic $k\in\mathbb{Q}$ (including $k=2$, $k=3$, and $k=\frac{557}{424}$), given by the quadratic equation in $Y=y^2$:*

$$(k^4+1)Y^2+6(x^2+k^2z^2)Y+\left(x^4-4x^3z-12x^2z^2-16xz^3-7z^4\right)=0.$$

Passing to the quotient by $y\mapsto -y$, the discriminant in $Y=y^2$ defines the ***genus-1 elliptic quotient curve*** *(smooth of genus 1 for generic $k\in\mathbb{Q}$ where the quartic discriminant in x/z is non-zero)*

$$V^2=(8-k^4)x^4+4(k^4+1)x^3z+6(2k^4+3k^2+2)x^2z^2+16(k^4+1)xz^3+(16k^4+7)z^4,$$

over which the genus-3 curve C_k is the double cover $(k^4+1)y^2=-3(x^2+k^2z^2)+V$, having rational points at $k=3$ (Row 1) and $k=\frac{557}{424}$ (Row 21).

15 The Opposite-Pair Surface ($e=-d$) and Its Sum-of-Fourth-Powers Identity

Theorem 15.1 (Opposite-Pair Surface Identities and Osculating Map)**.** *When $e=-d$ (so $d+e=0$ and $s=a+b+c=a-2x$, as in Row 1), the fourth-power equation $a^4+b^4+c^4+2d^4=(a+b+c)^4$ satisfies two equivalent identities:*

$$\left(y^2+3x^2\right)^2+d^4=4x(2x-a)(a^2-ax+2x^2), \tag{51}$$

$$\left(y^2+3x^2\right)^2+d^4+a^4=(4x^2-2ax+a^2)^2. \tag{52}$$

Expanding around Row 1 normalized at $(\frac{3}{4},-1,-1,-\frac{1}{2},\frac{1}{2})$ via $a=\frac{3}{4}x+a_2z$, $b=-x+y$, $c=-x-y$, $d=-\frac{1}{2}x+d_2z$, $e=-d$, the 6-point osculating parabola $y^2=z(\beta x+\gamma z)$ of Theorem 6.1 restricts on $e_2=-d_2$ to the exact homogeneous polynomials:

$$\begin{aligned}
\beta(a_2,d_2)&=-\frac{19a_2-2d_2}{24},\\
\gamma(a_2,d_2)&=\frac{1367a_2^2+76a_2d_2-868d_2^2}{3456},\\
R_1(a_2,d_2)&=-\frac{139915a_2^3+1290a_2^2d_2+16644a_2d_2^2-84680d_2^3}{41472},\\
R_0(a_2,d_2)&=-\frac{1868689a_2^4+207784a_2^3d_2-2367336a_2^2d_2^2-131936a_2d_2^3+12697360d_2^4}{11943936},
\end{aligned}$$

with $\frac{x}{z}=-\frac{R_0(a_2,d_2)}{R_1(a_2,d_2)}$ and $\left(\frac{y}{z}\right)^2=\frac{\gamma R_1-\beta R_0}{R_1}$ whenever $R_1(a_2,d_2)\neq 0$ (yielding a rational point on the surface whenever $\frac{\gamma R_1-\beta R_0}{R_1}$ is a rational square).

Proof. Setting $z=0$ and $w=d$ in the Three-Squares Identity (8) immediately gives $(y^2+3x^2)^2+d^4-(4x^2-2ax+a^2)^2=-a^4$. On the slice $a=\frac{3}{4}x+a_2z$, $b=-x+y$, $c=-x-y$, $d=-\frac{1}{2}x+d_2z$, $e=-d$, we have $s=-\frac{5}{4}x+a_2z$ and $x_0=1$, so expanding $\frac{1}{2}(s^4-a^4-b^4-c^4-2d^4)+(3x^2+y^2)^2$ in x and applying (24) yields the stated exact expressions for β,γ,R_1,R_0. □

16 The Equal-Pair Surface ($d=e$) and Dual Osculating Families

Proposition 16.1 (Structure of the Equal-Pair Surface $d=e$)**.** *When two coordinates are equal (satisfied in the census by* ***Row 1****: $s=5$, $(-3,4,4,2,-2)$ with equal pair $(4,4)$;* ***Row 2****: $s=15$, $(13,12,-6,-2,-2)$ with equal pair $(-2,-2)$; and* ***Row 5****: $s=135$, $(111,114,-58,-16,-16)$ with equal pair $(-16,-16)$):*

1. *Placing the equal pair in (d,e) so that $d=e$ fixes $r=\frac{u+3e}{2}$ in the Centered Double-Depressed-Cubic Identity (48), reducing the surface to*

$$2\left(y^2+3x^2\right)^2=(u+2e)^4-2e^4-uv(4v^2+u^2). \tag{53}$$

In the affine chart $d=e=-2$, Row 2 lies at $(a,x,y,u,v)=(13,-3,9,19,\frac{7}{2})$ and Row 5 (divided by 8) lies at $(a,x,y,u,v)=(\frac{111}{8},-\frac{7}{2},\frac{43}{4},\frac{167}{8},\frac{55}{16})$, with secant displacement satisfying $\Delta y=\frac{43}{4}-9=\frac{7}{4}=2(\frac{111}{8}-13)=2\Delta a$.

2. *Placing the equal pair in (b,c) so that $y_0=0$ and $x_0=1$ normalizes Row 1 to $(\frac{3}{4},-1,-1,-\frac{1}{2},\frac{1}{2})$, Row 2 to $(\frac{13}{2},-1,-1,6,-3)$, and Row 5 to $(\frac{111}{16},-1,-1,\frac{57}{8},-\frac{29}{8})$, providing three base points for the 2-parameter osculating-parabola map of Theorem 6.1.*

17 The Hyperbolic $(s-d)(s-e)$ Factorization on the Three-Squares Identity

Theorem 17.1 (Hyperbolic $(s-d)(s-e)$ Factorization)**.** *In the Three-Squares Identity (8), imposing the relation*

$$\left[4(x^2+xz+z^2)-2a(x+z)+a^2\right]-\left(w^2+3z^2\right)=2ax+2m^2$$

cancels $2ax$ and a^2 identically to yield the hyperbolic equation

$$(z+2x-a)^2-w^2=(s-d)(s-e)=2m^2, \tag{54}$$

which is solved rationally for all (a,x,m,r) $(r\neq 0)$ by

$$z+2x-a=\frac{r^2+2m^2}{2r},\qquad w=\frac{r^2-2m^2}{2r}\qquad \left(s-d=-r,\ s-e=-\frac{2m^2}{r}\right), \tag{55}$$

while the fourth-power equation reduces to the quadratic in z:

$$\left(y^2+3x^2\right)^2=(4mz+m(2x-a))^2+3m^2(2x-a)^2-4m^4-\frac{a^4+(2x-a)^4}{2}+8x^4. \tag{56}$$

Proof. Since $4(x^2+xz+z^2)-2a(x+z)+a^2-(w^2+3z^2)=(z+2x-a)^2-w^2+2ax$, equating to $2ax+2m^2$ gives $(z+2x-a)^2-w^2=2m^2$. Because $d=-z+w$, $e=-z-w$, and $s=a-2x-2z$, we have $s-d=a-2x-z-w=-(z+2x-a+w)=-r$ and $s-e=a-2x-z+w=-(z+2x-a-w)=-\frac{2m^2}{r}$, whose product is $(s-d)(s-e)=(-r)(-\frac{2m^2}{r})=2m^2$. Finally, writing $A=4(x^2+xz+z^2)-2a(x+z)+a^2$ and $B=w^2+3z^2=A-(2ax+2m^2)$, we have $A^2-B^2=4(ax+m^2)A-4(ax+m^2)^2$; subtracting $a(a^3-8axz+16xz(x+z))$ in (8) and collecting powers of z yields (56). □

18 Explicit Polynomial Parametrization for the $e_0=\frac{1}{3}$ Sector (Row 7)

Theorem 18.1 (Explicit Polynomial Parametrization and Infinite-Order Proof for the $e_0=\frac{1}{3}$ Sector)**.** *In Row 7 of Table I, $(a_0,b_0,c_0,d_0,e_0^*)=(-18,146,-121,182,16)$ with $s=205$ satisfies*

$$\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0}=\frac{-18+146-121}{-18+146-121+182}=\frac{7}{189}=\frac{1}{27}=\left(\frac{1}{3}\right)^3\implies e_0=\frac{1}{3}.$$

Setting $z=189$, $x=\frac{1}{2}X$, and $y=\frac{1}{2}Y$ on $H_{1/3}$ and parameterizing the resulting conic $(3X^2+Y^2)^2=237216X^2+103398400X+7789670896$ around $(X_0,Y_0,U_0)=(-25,267,73164)$ via $3X^2+Y^2=$

$73164+16u(X+25)$ *yields the explicit polynomial parametrization:*

$$\begin{aligned} a(u) &= -432u^2 - 219492u + 8981952, \\ b(u,W) &= \frac{3(200u^2+73164u-3045875)+3W}{2}, \\ c(u,W) &= \frac{3(200u^2+73164u-3045875)-3W}{2}, \\ d(u) &= 546(8u^2-7413), \\ e(u) &= 384u^2 - 73164u + 2504726, \end{aligned} \tag{57}$$

with exact coordinate sum

$$s(u) = a(u)+b(u,W)+c(u,W)+d(u)+e(u) = 4920u^2 - 73164u - 1698445, \tag{58}$$

where (u,W) lies on the smooth genus-1 curve

$$\mathcal{E}_{1/3}: \quad W^2 = -4802496u^4+278353600u^3-12403862688u^2+997806276600u-23811513788559, \tag{59}$$

whose quartic polynomial has non-zero discriminant

$$\Delta(\mathcal{E}_{1/3}) = -4182047393100605364421481477748279088349149354948783216420454 4 \neq 0,$$

with base point at the tangent slope

$$u_0 = \frac{204325}{5226}, \qquad W_0 = \frac{2926601209117}{2275923} = \frac{35119214509404}{5226^2}.$$

Under the integral shift $u = \frac{204325+\tilde{u}}{5226}$ and $\tilde{W} = 5226^2 W$, $\mathcal{E}_{1/3}$ has integer quartic coefficients

$$\begin{aligned} &c_4 = -4802496, && c_3 = -2470404067200, && c_2 = -650070819042012288, \\ &c_1 = 22304645772077402990400, && E_0 = 35119214509404, \end{aligned}$$

and Proposition 8.2 maps $\mathcal{E}_{1/3}$ isomorphically onto the integral Weierstrass cubic $E_{1/3}: Y_W^2 = X_W^3 + a_2X_W^2 + a_4X_W + a_6$ with

$$\begin{aligned} a_2 &= -2600283276168049152, \\ a_4 &= -5025388256211319449261981524237352 96, \\ a_6 &= -6569068853598446554666067815559658723261097468577710 08, \end{aligned}$$

on which the base point has non-integral coordinate

$$X_W(P_0^W) = \frac{23791919048547130292992}{7921} = \frac{23791919048547130292992}{89^2} \notin \mathbb{Z},$$

proving by the Nagell–Lutz Theorem that P_0^W ***has infinite order in*** *$\mathcal{E}_{1/3}(\mathbb{Q})$ (*$\operatorname{rank} E_{1/3}(\mathbb{Q}) \geq 1$*). Furthermore, $\Phi: \mathcal{E}_{1/3} \dashrightarrow V \subset \mathbb{P}^4$ has degree 1 onto its image curve, avoids zero coordinates outside at most 24 points on $\mathcal{E}_{1/3}$, and every rational point on $\mathcal{E}_{1/3}(\mathbb{Q})$ satisfies $u \equiv 0 \pmod 5$ (or $5 \mid \operatorname{den}(u)$), forcing $5 \mid s_{\text{prim}}$ and generating infinitely many primitive solutions in the $5 \mid s$ sector.*

Proof. This follows from Theorem 5.1 with $e_0 = \frac{1}{3}$, $z_0 = 189$, $X_0 = 2x_0 = -25$, $Y_0 = 2y_0 = 267$, and $t = 16u$. Because $237216 = 32 \cdot 7413$, the denominator $t^2 - \mathcal{A} = 256u^2 - 237216 = 32(8u^2 - 7413)$ shares a factor of 32 with the numerator of $X(u)$ and $U(u)$, which cancels to yield (57), (58), and (59). By (32), the tangent slope in t is $t_0 = \frac{2(237216)(-25)+103398400}{2(73164)} = \frac{1634600}{2613}$, so $u_0 = \frac{t_0}{16} = \frac{204325}{5226}$. By Theorem 5.1, Φ has degree 1 onto its image curve and $abcde = 0$ defines at most 24 points on $\mathcal{E}_{1/3}$. Modulo 5, the right-hand side of (59) takes the values $(1,2,3,3,2)$ for $u \equiv (0,1,2,3,4) \pmod 5$, so $u \equiv 0 \pmod 5$ for 5-integral u (giving $s(u) = 4920u^2 - 73164u - 1698445 \equiv 0 \pmod 5$, since $5 \mid 4920$ and $1698445 = 5 \cdot 339689$, while $a(u) \equiv 8981952 \equiv 2 \not\equiv 0 \pmod 5$), whereas if $5 \mid q$ for $u = p/q$ ($\gcd(p,q) = 1$) then $q^2 s(p/q) = 4920p^2 - 73164pq - 1698445q^2 \equiv 0 \pmod 5$ and $q^2 a(p/q) \equiv -432p^2 \equiv 3p^2 \not\equiv 0 \pmod 5$. □

19 Secant-Interpolation Cubic and Slope Map on the Paired-Solution Surface ($u = 2e$)

Theorem 19.1 (Secant Factorization and Rational Slope Map on $u = 2e$)**.** *In Proposition 13.1, substituting $V = 2X + 1$ (where $X = \frac{x}{e} = -\frac{b+c}{2e}$) brings the Paired-Solution Surface $u = 2e$ into the form*

$$2\left(Y^2+3X^2\right)^2 = 12R^3+27R-\left(64X^3+96X^2+64X+17\right), \tag{60}$$

on which Row 3 and Row 4 have opposite X-coordinates $X = \mp\frac{5}{8}$:

$$\textit{Row 3: } (R,X,Y)=\left(\frac{3}{8},-\frac{5}{8},\pm\frac{9}{8}\right), \qquad \textit{Row 4: } (R,X,Y)=\left(\frac{123}{8},+\frac{5}{8},\pm\frac{97}{8}\right).$$

1. ***Secant Interpolation:*** *Along the linear interpolant $R(X) = \frac{63}{8} + 12X$ and $W(X) = Y^2 + 3X^2 = \frac{1205}{16} + \frac{583}{5}X$ connecting Row 3 and Row 4, the cubic equation (60) factors completely over $\mathbb{Q}$ as*

$$\frac{1}{800}(8X-5)(8X+5)(258400X+169211)=0.$$

2. ***Tangent-Slope Parametrization of (R, X, Y^2) Around Row 3:*** *Because the right-hand side of (60) has degree 3 in (R, X) while the left-hand side is $2W^2$ (degree 2 in $W = Y^2 + 3X^2$), expanding around Row 3 via $X = -\frac{5}{8} + t$, $R = \frac{3}{8} + mt$, and $W = \frac{39}{16} + \frac{19(27m-16)}{156}t$ makes $t = 0$ a double root for every slope $m \in \mathbb{Q}$ (noting that $3m^3 - 16 \neq 0$ for all $m \in \mathbb{Q}$ since $\frac{16}{3} \notin (\mathbb{Q}^\times)^3$), yielding the rational parameterization of $(X(m), R(m), Y(m)^2)$:*

$$X(m) = -\frac{91260m^3-98901m^2+311904m-287104}{48672(3m^3-16)},$$
$$R(m) = \frac{3}{8}+\frac{m\left(98901m^2-311904m-199616\right)}{48672(3m^3-16)},$$

and $789654528(3m^3-16)^2Y(m)^2 = P_6(m)$, where

$$P_6(m) = 24824357064m^6-41251258152m^5-69076393785m^4-136173582528m^3 + 162207048960m^2+191717867520m+309338767360.$$

*Since $789654528 = 3\cdot 16224^2$ and the sextic polynomial $P_6(m)$ on the right-hand side is irreducible over $\mathbb{Q}$ with non-zero discriminant, a rational slope $m \in \mathbb{Q}$ lifts to a rational point $(R(m), X(m), Y(m))$ on (60) if and only if (m, y) is a rational point on the smooth **genus-2** hyperelliptic curve $y^2 = 3P_6(m)$ (which by Faltings' Theorem has only finitely many rational points; a search over $|p| \leq 600, 1 \leq q < 300$ finds no rational point $m = p/q$, and Row 4 is not in this family because the secant $W(X)$ between Rows 3 and 4 has slope $\frac{583}{5}$ rather than the tangent slope $\frac{19(27(12)-16)}{156} = \frac{1463}{39}$ at Row 3).*

Proof. Substituting $X = -\frac{5}{8} + t$ and $R = \frac{3}{8} + mt$ into the right-hand side of (60) divided by 2 gives $W_0^2 + c_1(m)t + c_2(m)t^2 + c_3(m)t^3$ with exact coefficients

$$W_0=\frac{39}{16},\quad c_1(m)=\frac{19(27m-16)}{32},\quad c_2(m)=\frac{3(9m^2+16)}{4},\quad c_3(m)=6m^3-32=2(3m^3-16)\neq 0$$

for all $m \in \mathbb{Q}$. Setting $W = W_0 + \alpha(m)t$ with $\alpha(m) = \frac{c_1(m)}{2W_0} = \frac{19(27m-16)}{156}$ cancels the t^0 and t^1 terms of $W^2 - \text{RHS} = 0$, leaving $t_{\text{new}}(m) = \frac{\alpha(m)^2-c_2(m)}{c_3(m)}$, which produces the stated rational functions for $X(m)$, $R(m)$, and $Y(m)^2 = W(m) - 3X(m)^2$. □

20 Classification of Algebraic Subfamilies and Complete 42-Row Census Index

We conclude with two summary tables: Table III synthesizes the special low-codimension algebraic surfaces and infinite parametric families developed in Sections 1–19, while Table IV records the all-integer canonical coordinates $(u,v,r,x,y,e)\in\mathbb{Z}^6$ of every primitive solution in Tables I and II on the Centered Double-Depressed-Cubic Threefold (Theorem 12.1(3)).

Intrinsic Memberships vs. Canonical Coordinate Labeling: Across the 42 primitive solutions in the census, the intrinsic (permutation-invariant) subvariety memberships are:

- **Bitangent Hyperplanes H_{e_0}:** $e_0=-1$ contains Rows $\{1,3,18\}$; $e_0=2$ contains Row $\{6\}$; $e_0=\frac{1}{3}$ contains Row $\{7\}$; and $e_0=\frac{3}{5}$ contains Row $\{40\}$ (Row 4 of Table II).
- **Special Algebraic Surfaces:** The Four-Squares Surface S_0 $(2a+b+c=0)$ contains Rows $\{1,21\}$; the Paired-Solution Surface $(s-d=3e)$ contains Rows $\{3,4,16,17\}$; the Equal-Pair Surface $(d=e)$ contains Rows $\{1,2,5\}$; and the Opposite-Pair Surface $(d=-e)$ contains Row $\{1\}$.

In Table IV, while $d_{\text{can}}=x_{\text{odd}}$ is canonically singled out by the 2-Adic Parity Law, the assignment of the four even coordinates to $(a_{\text{can}},b_{\text{can}},c_{\text{can}},e_{\text{can}})$ involves choosing which two even coordinates are placed in a_{can} and e_{can} (22 of the 181 distinct coordinate values in the census occur in two or more rows). All 42 rows of Table IV exactly reconstruct the solutions of Tables I–II and satisfy (48) in $\mathbb{Z}^6$.

Table III: Classification of Special Algebraic Surfaces and Parametric Subfamilies

Algebraic Surface / Family	**Defining Relation on** (a,b,c,d,e,s)	**Reduced Canonical Equation**	**Solutions in Census Belonging to Family**
Bitangent Hyperplane H_{-1} ($e_0=-1$)	$2(a+b+c)+d=0$	$(3x^2+y^2)^2=z[24wx^2-12w(w+z)x+(2w^3+3w^2z+2wz^2-8z^3)]$	**Row 1** ($s=5, z=1, w=2$ or $9z$) **Row 3** ($s=15, z=7, w=6$) **Row 18** ($s=515, z=169=13^2$, $w\in\{108,401,867=3\cdot 17^2\}$)
Bitangent Hyperplane H_2 ($e_0=2$)	$7(a+b+c)+8d=0$	$(3x^2+y^2)^2=z[48(2w-15z)x^2+8(3w^2+3wz-255z^2)x+C_2(w,z)]$	**Row 6** ($s=175, z=2, w=81, x=23, y=49$)
Bitangent Hyperplane $H_{1/3}$ ($e_0=\frac{1}{3}$)	$26(a+b+c)-d=0$	$(3X^2+Y^2)^2=237216X^2+103398400X+7789670896$	**Row 7** ($s=205, z=189, X=-25, Y=267$)
Bitangent Hyperplane $H_{3/5}$ ($e_0=\frac{3}{5}, 5\nmid s$)	$98(a+b+c)-27d=0$	$(3X^2+Y^2)^2=147744X^2+21628032X+221539312$	**Row 4 of Table II** (Row 40, $s=551, X=9, Y=143$)
Four-Squares Surface S_0	$2a+b+c=0$ ($a=x$)	$(y^2+3x^2)^2+(w^2+3z^2)^2+x^2(x+4z)^2=(3x^2+2xz+4z^2)^2$	**Row 1** ($s=5, (-3,1,-1,3)$) **Row 21** ($s=625$, proj. rep. $2\times$: $(112,424,-681,557)$)
Paired-Solution Surface ($u=2e$)	$s-d=3e$ ($a+b+c=2e$)	$2(Y^2+3X^2)^2=12R^3+27R-(64X^3+96X^2+64X+17)$	**Row 3 & Row 4** ($s=15,135, e=8$) **Row 16 & Row 17** ($s=435,515, e=104$)
Non-Divisible Paired Slice ($5\nmid s$)	$s-d=625, e=-70$	$2(y^2+3x^2)^2=625r(4r^2+625^2)-695v(4v^2+695^2)-70^4$	**Row 2 & Row 3 of Table II** (Rows 38, 39; $s=413,507$)
Opposite-Pair Surface	$d+e=0$ ($e=-d$)	$(y^2+3x^2)^2+d^4+a^4=(4x^2-2ax+a^2)^2$	**Row 1** ($s=5: 48^2+2^4+3^4=49^2$)
Equal-Pair Surface	$d=e$ ($w=0$)	$2(y^2+3x^2)^2=(u+2e)^4-2e^4-uv(4v^2+u^2)$	**Row 1** ($s=5$, pair $4,4$) **Row 2** ($s=15, d=e=-2$) **Row 5** ($s=135, d=e=-16$)

Table IV: Complete 42-Row All-Integer Canonical Coordinates (Theorem 12.1) and Parametric Classification

For each row of Tables I and II, placing the unique odd coordinate x_{odd} in d_{can} and the four even coordinates in $(a_{\text{can}}, b_{\text{can}}, c_{\text{can}}, e_{\text{can}})$ yields all-integer centered coordinates $(u, v, r, x, y, e) \in \mathbb{Z}^6$ satisfying

$$2(y^2+3x^2)^2 = (u+e)r(4r^2+(u+e)^2) - uv(4v^2+u^2) - e^4.$$

Row	s	d_{can}	a_{can}	u	v	r	x	y	e_{can}	Osculating Map & Subvariety / Canonical Labeling Notes
1	5	−3	2	10	−3	1	−4	0	−2	Thm. 6.1 ($b=c=4$); H_{-1} (Thm. 1.1), S_0 (Thm. 2.2), $d=-e$ (Thm. 15.1)
2	15	13	−2	4	−4	14	−3	9	−2	Thm. 6.1 ($d=e$); Equal-Pair Surface $d=e=-2$ (Prop. 16.1)
3	15	−9	6	16	−2	3	−5	9	8	Thm. 10.1; H_{-1} ($e_0=-1$) & Paired Surface $u=2e$ (Prop. 13.1, Thm. 19.1)
4	135	111	26	16	18	123	5	−97	8	Thm. 10.1; Paired Surface $u=2e$ ($e=8$, Thm. 7.3, Prop. 13.1, Thm. 19.1)
5	135	111	−16	40	−36	123	−28	86	−16	Thm. 6.1 ($d=e$); Equal-Pair Surface $d=e=-16$ (Prop. 16.1)
6	175	173	26	16	18	174	5	−67	−14	Thm. 10.1; Bitangent Hyperplane H_2 ($e_0=2$, Thm. 5.1, Ex. 5.2)
7	205	−121	−18	310	−173	42	−164	18	16	Thm. 10.1; Bitangent Hyperplane $H_{1/3}$ ($e_0=\frac{1}{3}$, Thm. 18.1)
8	225	57	−104	266	−237	141	−185	1	−98	Thm. 10.1; Centered Canonical Fiber $(u,e)=(266,-98)$ (Thm. 12.1)
9	235	73	126	186	33	154	−30	186	−24	Thm. 10.1; Centered Canonical Fiber $(u,e)=(186,-24)$ (Thm. 12.1)
10	295	143	124	78	85	219	23	−239	74	Thm. 10.1; Iso-Sum Pair $s=295$ with Row 11 (Thm. 12.1)
11	295	277	74	−20	84	286	47	−151	38	Thm. 10.1; Iso-Sum Pair $s=295$ with Row 10; $e_{\text{can}}=38$ with Row 12
12	315	−253	98	530	−167	31	−216	48	38	Thm. 10.1; Labeling $e_{\text{can}}=38$ with Row 11 (Thm. 11.1)
13	325	111	96	148	22	218	−26	268	66	Thm. 10.1; Shared $d_{\text{can}}=111$ with Rows 4, 5 (Thm. 12.1)
14	355	−271	96	618	−213	42	−261	37	8	Thm. 10.1; Labeling $e_{\text{can}}=8$ with Rows 3, 4 ($a_{\text{can}}=96$ with Row 13)
15	425	−63	−306	350	−481	181	−328	16	138	Thm. 10.1; Centered Canonical Fiber $(u,e)=(350,138)$ (Thm. 12.1)
16	435	123	194	208	90	279	−7	361	104	Thm. 10.1; Paired Surface $u=2e$ ($e=104$, Thm. 7.3, Prop. 13.1)
17	515	203	144	208	40	359	−32	426	104	Thm. 10.1; Paired Surface $u=2e$ ($e=104$, Thm. 7.3, Prop. 13.1)
18	515	−407	338	1036	−180	54	−349	3	−114	Thm. 10.1; Bitangent Hyperplane H_{-1} ($e_0=-1, z=169=13^2, w\in\{108,401,867\}$)
19	585	97	−332	614	−639	341	−473	41	−126	Thm. 10.1; Centered Canonical Fiber $(u,e)=(614,-126)$ (Thm. 12.1)
20	625	497	76	132	10	561	−28	460	−4	Thm. 10.1; Iso-Sum Pair $s=625=5^4$ with Row 21 (Thm. 12.1)
21	625	619	62	−50	87	622	56	−212	56	Thm. 10.1; Four-Squares Surface S_0 ($2a+b+c=0$, Thm. 2.2, Prop. 14.1)
22	645	317	116	230	1	481	−57	525	98	Thm. 10.1; Iso-Sum Pair $s=645$ with Row 23 (Thm. 12.1)
23	645	637	64	14	57	641	25	−251	−6	Thm. 10.1; Iso-Sum Pair $s=645$ with Row 22 (Thm. 12.1)
24	685	37	−252	794	−649	361	−523	141	−146	Thm. 10.1; Centered Canonical Fiber $(u,e)=(794,-146)$ (Thm. 12.1)
25	705	−447	128	1204	−474	129	−538	106	−52	Thm. 10.1; Iso-Sum Pair $s=705$ with Row 26 (Thm. 12.1)
26	705	667	54	10	49	686	22	−394	28	Thm. 10.1; Iso-Sum Pair $s=705$ with Row 25 (Thm. 12.1)
27	715	681	28	32	12	698	−2	390	2	Thm. 10.1; Centered Canonical Fiber $(u,e)=(32,2)$ (Thm. 12.1)
28	755	289	−108	378	−297	522	−243	505	88	Thm. 10.1; Labeling $e_{\text{can}}=88$ with Row 29 (Thm. 11.1)
29	775	−657	322	1344	−350	59	−511	93	88	Thm. 10.1; Labeling $e_{\text{can}}=88$ with Row 28 (Thm. 11.1)
30	835	−117	502	456	274	359	23	−651	496	Thm. 10.1; Iso-Sum Pair $s=835$ with Row 31 (Thm. 12.1)
31	835	587	474	570	189	711	−48	620	−322	Thm. 10.1; Iso-Sum Pair $s=835$ with Row 30 (Thm. 12.1)
32	885	429	488	234	371	657	127	−681	222	Thm. 10.1; Centered Canonical Fiber $(u,e)=(234,222)$ (Thm. 12.1)
33	905	−583	202	1550	−573	161	−674	160	−62	Thm. 10.1; Centered Canonical Fiber $(u,e)=(1550,-62)$ (Thm. 12.1)
34	945	811	−176	2	−177	878	−89	635	132	Thm. 10.1; Centered Canonical Fiber $(u,e)=(2,132)$ (Thm. 12.1)
35	965	−697	264	1754	−613	134	−745	39	−92	Thm. 10.1; Centered Canonical Fiber $(u,e)=(1754,-92)$ (Thm. 12.1)
36	985	291	−426	1086	−969	638	−756	180	−392	Thm. 10.1; Centered Canonical Fiber $(u,e)=(1086,-392)$ (Thm. 12.1)
37	363	315	−262	18	−271	339	−140	90	30	Thm. 10.1; $5\nmid s$ Sector ($s-a_{\text{can}}=5^4$, Thms. 3.1, 12.1)
38	413	55	−212	428	−426	234	−320	70	−70	Thm. 10.1; $5\nmid s$ Paired Slice ($s-a_{\text{can}}=5^4, e=-70$, Sec. 13)
39	507	395	−118	182	−209	451	−150	300	−70	Thm. 10.1; $5\nmid s$ Paired Slice ($s-a_{\text{can}}=5^4, e=-70$, Sec. 13)
40	551	335	−74	36	−92	443	−55	435	180	Thm. 10.1; Bitangent Hyperplane $H_{3/5}$ ($e_0=\frac{3}{5}, 5\nmid s$, Thm. 9.1)
41	967	−465	342	1842	−579	251	−750	150	−410	Thm. 10.1; $5\nmid s$ Sector ($s-a_{\text{can}}=5^4$, Thms. 3.1, 12.1)
42	991	−259	500	840	80	366	−170	760	410	Thm. 10.1; $5\nmid s$ Sector ($s-d_{\text{can}}=2\cdot 5^4$, $10\mid a_{\text{can}}, b, c, e$, Thm. 3.1)

A Reproducible C99 Verification Code for the $0 < s < 1000$ Census

To make the completeness of the 42-row census in Proposition 0.1(3) and Tables I–II independently verifiable, we provide two complete, self-contained C99 programs:

1. **Meet-in-the-Middle Even-Coordinate Enumerator (`census_verify.c`, Listing 1):** Implements the two-sum bucket decomposition of Proposition 0.1(3). Compiling and executing via

   ```
   gcc -O3 -std=c99 census_verify.c -o census_verify -lm && ./census_verify
   ```

   constructs all $\binom{998+1}{2} = 498{,}501$ ordered non-zero pairs (B, C) $(-499 \le C \le B \le 499)$ across the 1,997 two-sum buckets $S_2 = B + C \in [-998, 998]$, explicitly verifies during insertion that zero duplicate $P_2 = B^4 + C^4$ values occur in any bucket (`DUP_P2=0`, with maximum bucket occupancy `MAX_BUCKET=499` of 500), and executes the two-pointer scan across all 249,500 odd pairs (s, x_{odd}) (124,750,000 loop iterations over $(s, x_{\text{odd}}, S_{BC})$, of which 124,749,997 scan non-empty bucket pairs), outputting the 42 primitive solutions and summary statistics in Listing 2.

2. **Unrestricted Fourth-Power-Budget Enumerator (`indep_census.c`, Listing 3):** An independent algorithm that orders $|x_1| \ge |x_2| \ge |x_3| \ge |x_4| \ge |x_5| \ge 1$, loops over x_1, x_2, x_3 within fourth-power-budget bounds, and solves $x_4 + x_5 = \sigma$, $x_4^4 + x_5^4 = R$ in closed form via $\sigma^4 + R = 2j^2$ and $(x_4 - x_5)^2 = 4j - 3\sigma^2$, **using no parity or mod-5 pruning and testing all $1 \le s \le 999$ including even s.** It finds 408 total non-zero solutions (42 of which are primitive, identical as a set to Tables I–II, and 0 with even s).

Listing 1: Complete C99 meet-in-the-middle verification program (`census_verify.c`).

```c
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>

typedef struct { int B, C; int64_t P2; } Pair;

int gcd_int(int a, int b) {
    if (a < 0) a = -a;
    if (b < 0) b = -b;
    while (b != 0) { int t = a % b; a = b; b = t; }
    return a;
}

static Pair buckets[1997][500];
static int b_len[1997];

int main(void) {
    int total_pairs = 0, count_div5 = 0, count_ndiv5 = 0, prev_max_B = 0;
    int dup_p2 = 0, max_bucket = 0;
    int64_t total_loop_iters = 0, nonempty_scans = 0;

    for (int s = 1; s < 1000; s += 2) {
        int max_B = (s - 1) / 2;
        if (max_B > prev_max_B) {
            int M = max_B;
            for (int B = -M; B <= M; B++) {
                if (B == 0) continue;
                for (int C = -M; C <= B; C++) {
                    if (C == 0 || (abs(B) != M && abs(C) != M)) continue;
                    int id = (B + C) + 998;
                    int64_t P2 = (int64_t)B*B*B*B + (int64_t)C*C*C*C;
                    int k = b_len[id]++;
                    if (b_len[id] > max_bucket) max_bucket = b_len[id];
                    while (k > 0 && buckets[id][k-1].P2 > P2) {
                        buckets[id][k] = buckets[id][k-1];
                        k--;
                    }
                    if (k > 0 && buckets[id][k-1].P2 == P2) dup_p2++;
                    buckets[id][k] = (Pair){B, C, P2};
```

```
                    total_pairs++;
                }
            }
            prev_max_B = max_B;
        }
        int64_t s4 = (int64_t)s * s * s * s;
        for (int xodd = -(s - 2); xodd <= (s - 2); xodd += 2) {
            if (xodd == 0) continue;
            int64_t x4 = (int64_t)xodd * xodd * xodd * xodd;
            int64_t target_P = (s4 - x4) >> 4;
            int target_S = (s - xodd) >> 1;
            int min_S_BC = (target_S + 1) >> 1;
            int max_S_BC = 2 * max_B;
            if (target_S + 2 * max_B < max_S_BC) max_S_BC = target_S + 2 * max_B;

            for (int S_BC = min_S_BC; S_BC <= max_S_BC; S_BC++) {
                total_loop_iters++;
                int S_DE = target_S - S_BC;
                int b1 = S_BC + 998, b2 = S_DE + 998;
                int n1 = b_len[b1], n2 = b_len[b2];
                if (n1 == 0 || n2 == 0) continue;
                nonempty_scans++;
                const Pair *P1 = buckets[b1], *P2 = buckets[b2];
                if (P1[0].P2 + P2[0].P2 > target_P || P1[n1-1].P2 + P2[n2-1].P2 < target_P) continue;
                int i = 0, j = n2 - 1;
                while (i < n1 && j >= 0) {
                    int64_t sumP = P1[i].P2 + P2[j].P2;
                    if (sumP < target_P) i++;
                    else if (sumP > target_P) j--;
                    else {
                        int B = P1[i].B, C = P1[i].C, D = P2[j].B, E = P2[j].C;
                        if (C >= D) {
                            int g = gcd_int(xodd, gcd_int(2*B, gcd_int(2*C, gcd_int(2*D, 2*E))));
                            if (g == 1) {
                                if (s % 5 == 0) count_div5++; else count_ndiv5++;
                                printf("SOL s=%3d xodd=%4d evens=(%4d,%4d,%4d,%4d)\n",
                                       s, xodd, 2*B, 2*C, 2*D, 2*E);
                            }
                        }
                        i++; j--;
                    }
                }
            }
        }
    }
    printf("SUMMARY: TOTAL_PAIRS=%d DUP_P2=%d MAX_BUCKET=%d LOOP_ITERS=%lld NONEMPTY_SCANS=%lld DIV5=%d NDIV5=%d
     TOTAL_SOL=%d\n",
           total_pairs, dup_p2, max_bucket, (long long)total_loop_iters, (long long)nonempty_scans,
           count_div5, count_ndiv5, count_div5 + count_ndiv5);
    return 0;
}
```

Listing 2: Exact 42-solution stdout log and output digests produced by `census_verify.c`. (The SHA-256 digest of the printed `SOL` lines is `ca627f4c...` via `./census_verify | grep "^SOL" | sha256sum` with standard trailing newline, or `2e5691f9...` when the 42 lines are joined by `"\n"` with no trailing newline; this digest verifies exact reproduction of the printed output.)

```
SOL s=  5 xodd=  -3 evens=(   4,   4,   2,  -2)
SOL s= 15 xodd=  -9 evens=(  14,   8,   6,  -4)
SOL s= 15 xodd=  13 evens=(  12,  -2,  -2,  -6)
SOL s=135 xodd= 111 evens=( 114, -16, -16, -58)
SOL s=135 xodd= 111 evens=(  92,  26,   8,-102)
SOL s=175 xodd= 173 evens=(  62,  26, -14, -72)
SOL s=205 xodd=-121 evens=( 182, 146,  16, -18)
SOL s=225 xodd=  57 evens=( 186, 184, -98,-104)
SOL s=235 xodd=  73 evens=( 216, 126, -24,-156)
SOL s=295 xodd= 143 evens=( 216, 124,  74,-262)
SOL s=295 xodd= 277 evens=( 104,  74,  38,-198)
SOL s=315 xodd=-253 evens=( 264, 168,  98,  38)
SOL s=325 xodd= 111 evens=( 294,  96,  66,-242)
SOL s=355 xodd=-271 evens=( 298, 224,  96,   8)
SOL s=363 xodd= 315 evens=( 230,  50,  30,-262)
SOL s=413 xodd=  55 evens=( 390, 250, -70,-212)
SOL s=425 xodd= -63 evens=( 344, 312, 138,-306)
```

```
SOL s=435 xodd= 123 evens=( 368, 194, 104,-354)
SOL s=507 xodd= 395 evens=( 450, -70,-118,-150)
SOL s=515 xodd=-407 evens=( 352, 346, 338,-114)
SOL s=515 xodd= 203 evens=( 458, 144, 104,-394)
SOL s=551 xodd= 335 evens=( 490, 180, -74,-380)
SOL s=585 xodd=  97 evens=( 514, 432,-126,-332)
SOL s=625 xodd= 497 evens=( 488,  76,  -4,-432)
SOL s=625 xodd= 619 evens=( 156,  62,  56,-268)
SOL s=645 xodd= 317 evens=( 582, 116,  98,-468)
SOL s=645 xodd= 637 evens=( 226,  64,  -6,-276)
SOL s=685 xodd=  37 evens=( 664, 382,-146,-252)
SOL s=705 xodd=-447 evens=( 644, 432, 128, -52)
SOL s=705 xodd= 667 evens=( 372,  54,  28,-416)
SOL s=715 xodd= 681 evens=( 392,  28,   2,-388)
SOL s=755 xodd= 289 evens=( 748,  88,-108,-262)
SOL s=775 xodd=-657 evens=( 604, 418, 322,  88)
SOL s=835 xodd=-117 evens=( 628, 502, 496,-674)
SOL s=835 xodd= 587 evens=( 668, 474,-322,-572)
SOL s=885 xodd= 429 evens=( 554, 488, 222,-808)
SOL s=905 xodd=-583 evens=( 834, 514, 202, -62)
SOL s=945 xodd= 811 evens=( 724, 132,-176,-546)
SOL s=965 xodd=-697 evens=( 784, 706, 264, -92)
SOL s=967 xodd=-465 evens=( 900, 600, 342,-410)
SOL s=985 xodd= 291 evens=( 936, 576,-392,-426)
SOL s=991 xodd=-259 evens=( 930, 500, 410,-590)
SUMMARY: TOTAL_PAIRS=498501 DUP_P2=0 MAX_BUCKET=499 LOOP_ITERS=124750000
         NONEMPTY_SCANS=124749997 DIV5=36 NDIV5=6 TOTAL_SOL=42
SHA-256 (trailing newline, grep "^SOL" | sha256sum):
  ca627f4c88ee43e0f6d0b14a185cf08ecc770d0ca20e4f52c32fcc83d43c7420
SHA-256 (joined by "\n", no trailing newline):
  2e5691f94c18ec9f9a5e1226783fc3d71e5cac83caa48eaa76508fcbda6aeb76
```

Listing 3: Independent parity-unrestricted fourth-power-budget search (`indep_census.c`).

```
// indep_census.c -- independent exhaustive search for a^4+b^4+c^4+d^4+e^4 = (a+b+c+d+e)^4 with all coordinates
     nonzero.
// Order |x1|>=|x2|>=|x3|>=|x4|>=|x5|>=1; loop x1,x2,x3 within fourth-power-budget bounds; solve the last pair in
     closed form:
//   x4+x5=sg, x4^4+x5^4=R  <=>  sg^4+R = 2 j^2  and (x4-x5)^2 = 4j-3 sg^2.
// Build: gcc -O3 -march=native -std=c99 indep_census.c -o indep_census -lm
// Run  : ./indep_census 1 999       (all s incl. even; prints "s gcd x1 x2 x3 x4 x5"; gcd==1 <=> primitive)
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
typedef long long ll;
static ll p4[4096]; static unsigned char q64[64],q63[63],q65[65],q11[11];
static inline int isq(ll n,ll*r){ if(n<0)return 0; ll k=(ll)sqrtl((long double)n); while(k*k>n)k--; while((k+1)*(k
     +1)<=n)k++; *r=k; return k*k==n; }
static inline int maybe_sq(ll w){ return q64[w&63]&&q63[w%63]&&q65[w%65]&&q11[w%11]; }
static int lo_bound(ll R,int mult){ int n=(int)ceil(pow((double)R/mult,0.25)); if(n<1)n=1; while(n>1&&(ll)mult*p4[
     n-1]>=R)n--; while((ll)mult*p4[n]<R)n++; return n; }
static int hi_bound(ll R){ if(R<1)return 0; int n=(int)floor(pow((double)R,0.25)); while(p4[n+1]<=R)n++; while(n
     >0&&p4[n]>R)n--; return n; }
static ll gcdl(ll a,ll b){ if(a<0)a=-a; if(b<0)b=-b; while(b){ll t=a%b;a=b;b=t;} return a; }
int main(int argc,char**argv){
  int smin=atoi(argv[1]), smax=atoi(argv[2]);
  for(int n=0;n<4096;n++) p4[n]=(ll)n*n*n*n;
  for(int i=0;i<64;i++)q64[(i*i)&63]=1; for(int i=0;i<63;i++)q63[(i*i)%63]=1;
  for(int i=0;i<65;i++)q65[(i*i)%65]=1; for(int i=0;i<11;i++)q11[(i*i)%11]=1;
  for(int s=smin;s<=smax;s++){            // ALL s (even too), no parity / mod-5 pruning
    ll S4=p4[s];
    for(int a1=lo_bound(S4,5);a1<=s-1;a1++) for(int g1=-1;g1<=1;g1+=2){ int x1=g1*a1;
     ll R1=S4-p4[a1]; if(R1<4)continue;
     int lo2=lo_bound(R1,4),hi2=hi_bound(R1-3); if(hi2>a1)hi2=a1;
     for(int a2=lo2;a2<=hi2;a2++) for(int g2=-1;g2<=1;g2+=2){ int x2=g2*a2;
      ll R2=R1-p4[a2]; int lo3=lo_bound(R2,3),hi3=hi_bound(R2-2); if(hi3>a2)hi3=a2; ll base=s-x1-x2;
      for(int a3=lo3;a3<=hi3;a3++) for(int g3=-1;g3<=1;g3+=2){ int x3=g3*a3;
        ll sg=base-x3, R=R2-p4[a3], sg2=sg*sg, V=R+sg2*sg2;
        if(V&1)continue; ll W=V>>1; if(!maybe_sq(W))continue; ll j; if(!isq(W,&j))continue;
        ll d2=4*j-3*sg2; if(d2<0)continue; ll m; if(!isq(d2,&m))continue; if((sg+m)&1)continue;
        ll x4=(sg+m)/2,x5=(sg-m)/2; if(!x4||!x5)continue; if(llabs(x4)>a3||llabs(x5)>a3)continue;
        if(p4[llabs(x4)]+p4[llabs(x5)]!=R)continue;
        printf("%d %lld %d %d %d %lld %lld\n",s,gcdl(gcdl(gcdl(x1,x2),gcdl(x3,x4)),x5),x1,x2,x3,x4,x5);
      }}}
  }
  return 0; }
```

References

[1] L. E. Dickson, *History of the Theory of Numbers, Vol. II: Diophantine Analysis*, Carnegie Institution of Washington, 1920.

[2] L. J. Lander, T. R. Parkin, and J. L. Selfridge, "A survey of equal sums of like powers," *Mathematics of Computation*, vol. 21, no. 99, pp. 446–459, 1967.

[3] N. D. Elkies, "On $A^4 + B^4 + C^4 = D^4$," *Mathematics of Computation*, vol. 51, no. 184, pp. 825–835, 1988.

[4] A. Choudhry, "On equal sums of fourth powers," *Rocky Mountain Journal of Mathematics*, vol. 30, no. 3, pp. 843–852, 2000.

[5] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed., Graduate Texts in Mathematics, vol. 106, Springer, 2009.